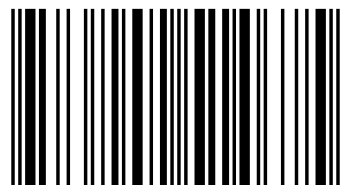


The practical Manual of Invertebrate and Chordate has been prepared to serve the special purpose of college students. This Manual explains the details of practical experiments and its needed explanation. The Manual can also be used for revision of practical techniques in the life time at University, College and Research level. Even this Manual can be used to U.G and P.G college students who are looking for a quick revision before the Practical Examinations. This Manual supports the college students to get the confidence and also reference just before attempting the university practical examination. This Manual has been written to give insight to not only to college students but also College Teachers as there are no such book which cover suitable original pictures and labelled diagram with needed explanation of invertebrate and chordate animal groups.



Mr. S.KALAIMANI is an expert in biological research in various fields of zoological studies, who is working as a Assistant Professor of Zoology, J.K.K.Nataraja college of Arts and Science, Kumarapalayam, Namakkal, Tamil Nadu, India. He has 14 years of experiences as a PG teacher of Biology in various school and college with diverse Research Experience.



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The Practical Manual of Invertebrate and Chordates

 **LAMBERT**
Academic Publishing

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THE PRACTICAL MANNUAL OF INVERTEBRATE AND CHORDATA

ZOOLOGY - PRACTICAL - I

INVERTEBRATES AND CHORDATES

(Affiliated Madurai kamaraj University syllabus -2017 onwards)

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“THE PRACTICAL MANNUAL OF INVERTEBRATE AND CHORDATA”

FOR AUTHOR USE ONLY

By

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Mr. S. KALAIMANI

Dr. C.KANDEEPAN

“THE PRACTICAL MANNUAL OF INVERTEBRATE AND CHORDATA”

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THE MAJOR ANIMALS PHYLA

INVERTEBRATES (ANIMALS WITHOUT BACKBONES)

Phylum: Porifera [Sponges]

Phylum: Cnidaria [Jellyfish & Corals]

Phylum: Platyhelminthes [Flatworms]

Phylum: Nematoda [Roundworms]

Phylum: Rotifera [wheel animals]

Phylum: Mollusca [Molluscs]

Phylum: Annelida [Segmented Worms]

Phylum: Arthropoda [Arthropods]

Trilobita

Chelicerata (spiders, scorpions)

Crustacea [lobster, crab, crayfish, shrimp]

Uniramia [centipedes, millipedes, insects]

Phylum: Echinodermata [Echinoderms]

CHORDATA (ANIMALS WITH BACKBONES)

Phylum: Chordata [Chordates]

Agnatha (lampreys, hagfish)

Chondrichthyes (sharks and rays)

Osteichthyes (bony fish)

Amphibia (frogs, salamanders)

Reptilia (snakes, lizards, turtles)

Aves (birds)

Mammalia (mammals)

GENERAL PROCEDURE OF ANIMAL COLLECTION

We are making a small animal collection consisting of 5 different animals, from 5 different phyla or subphyla. You will NOT get these items back so please don't include a keepsake or treasure that you want to hold onto. ACC will provide nets, vials, jars, preservative, insect pins, some labels, etc.

THE GOALS FOR THIS COLLECTION:

- ❖ Learn to visually recognize some of the animals common to the area
- ❖ Learn something about the ecology and behavior of these animals
- ❖ Learn how to use identification manuals and keys
- ❖ Become acquainted with the taxonomy and classification of animals, and
- ❖ Learn how to properly preserve and label museum specimens

You will be given more information on proper preservation and presentation of specimens in lab. You might also want to consult some Identification keys for the kinds of animals that you are interested in collecting. The animal collection will be worth 50 points and is due the Monday of the last week of classes.

GRADE FOR THE COLLECTION WILL BE BASED ON THE FOLLOWING CRITERIA:

- ❖ Diversity and originality of your collection
- ❖ Ability to follow correct procedures for preserving and displaying specimens as described in the materials
- ❖ Quality of the preservation technique
- ❖ Accuracy of identification (usually to species) & common name

You do not need to kill anything to make this collection. Some examples of the kinds of collections you can make: soil and leaf litter organism aquatic organisms shells skeletons and/or skulls nests, burrows, tunnels, etc., plaster casts of footprints or tracks photographs parasites fossils

PERMANENT SLIDES:

permanent slides are used for small or microscopic animals 70% alcohol/10% formalin: is used for most invertebrates including insect larvae, larger invertebrates and some vertebrates will need to be injected Study Skins: for birds and mammals Dried Specimens: for bones, shells, nests, etc., Pinned Specimens: adult insects are generally dried on insect pins and mounted in an insect box .Each and every specimen must be properly labeled. The type and location of the label depends on the kind of collection and preservation that you do.

ALL LABELS MUST INCLUDE THE FOLLOWING INFORMATION:

1. Collection Locality (including nearest city)
2. Date of collection General Habitat (e.g. woods, pond, soil, tree bark, grass, etc.,)
3. Scientific Name & Common Name Your Name

ZOOLOGY ANIMAL COLLECTION AND TECHNIQUES

- ❖ We can provide containers, vials or bags along with preservatives for your collection.
- ❖ We have several dozen identification guides for many kinds of central Texas animals. You can also find ID guides at the PIN Library and most public libraries in Austin. You can also find quite a lot of information by searching the web
- ❖ More information on proper collecting and preserving techniques are in a binder in the lab room
- ❖ More detailed Information on preserving various kinds of specimens:

A. ANIMALS COLLECTED LIVE:

- ❖ Most invertebrates except those listed below:
- ❖ Kill and preserve in jars of 10% formalin solution or 70% alcohol (ethyl or isopropyl); make label with India ink and place inside jar
- ❖ Most insects except butterflies, moths and dragonflies:
- ❖ Kill in killing jar or freezer; pin properly; allow to dry; label should be no larger than 1"x1/2" and placed on pin below insect
- ❖ Butterflies, moths and dragonflies:
- ❖ Kill in killing jar or freezer; pin and arrange wings on spreading board or piece of Styrofoam; allow to dry; label should be no larger than 1"x1/2" and placed on pin below insect
- ❖ Vertebrates except for birds and mammals (e.g.: fish, frogs, salamanders, lizards, snakes, turtles) kill in freezer; fix and preserve in jar of 10% formalin solution; use syringe to inject formalin solution into the body cavity in several places; make label with India ink and place inside jar
- ❖ Birds and Mammals:

B. ANIMALS COLLECTED DEAD BUT IN GOOD CONDITION:

Most invertebrates and lower vertebrates:

- ❖ Preserve in jars of 10% formalin solution or 70% alcohol; label as above

- ❖ Insects: If they are freshly dead, use above procedures; if they are long dead and dried out you will need to place them in a "relaxing chamber" to soften them up so they can be pinned and arranged; label as above
- ❖ Birds and Mammals:
- ❖ "Road kills" in good condition can be skinned and study skins prepared as above
- ❖ Skull or Skeletal mounts:
- ❖ Flesh can be removed by several means such as boiling, staking near fire ant hill or using demisted beetles; once bones are deflated they can be placed in a bleach or hydrogen peroxide solution to whiten; allow to dry; place in bag or box with complete label tied to skull if possible

C. NESTS, FEATHERS, EGGS, RACKS, SHELLS, FEATHERS:

- ❖ Clean up as much as possible and place in bag or box with complete label

D. FOOTPRINTS AND BURROWS:

- ❖ Use plaster of Paris to pour into print, allow to set then remove and clean off; place in bag or box with label

E. PHOTOGRAPHS:

- ❖ Should be original photographs; most 'good' photographs will need a telephoto lens for vertebrates or close-up lenses for small invertebrates and insects - the specimen should take up a substantial part of the animals.



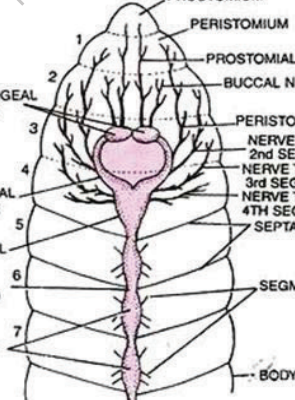
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DISSECTION CHARTS

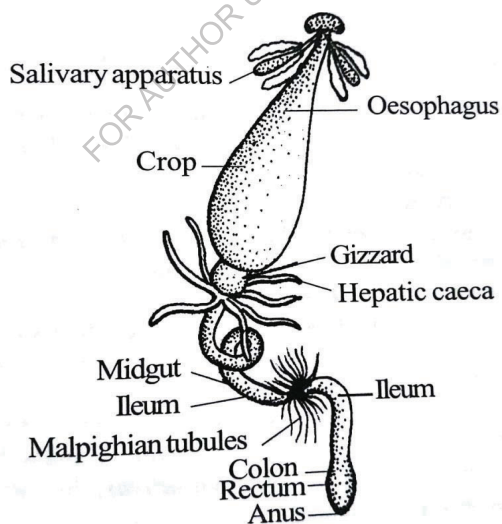
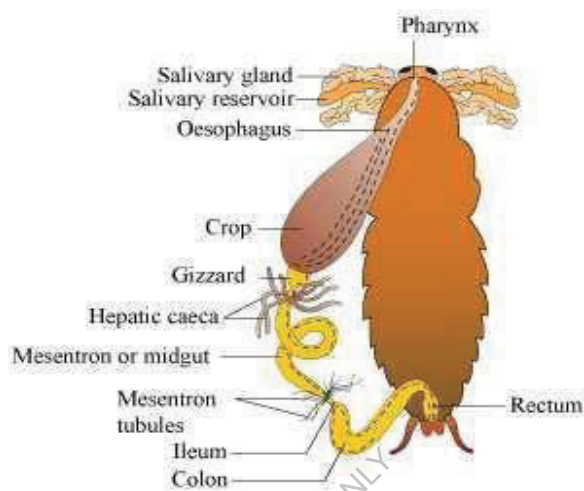
(ANIMAL NAME AND ITS DISSECTION DIAGRAMS)

nerve

PROSTOMIUM



COCKROACH – DIGESTIVE SYSTEM



COCKROACH-NERVOUSE SYSTEM

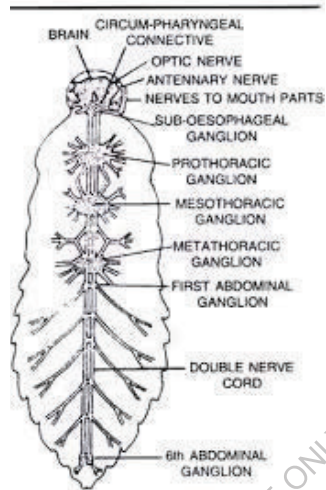
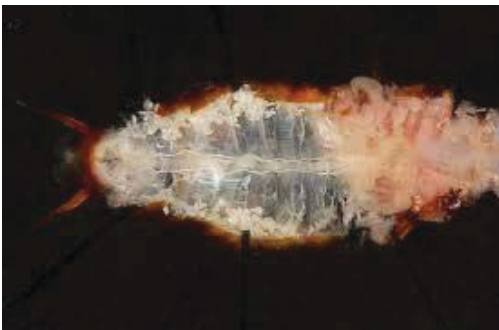
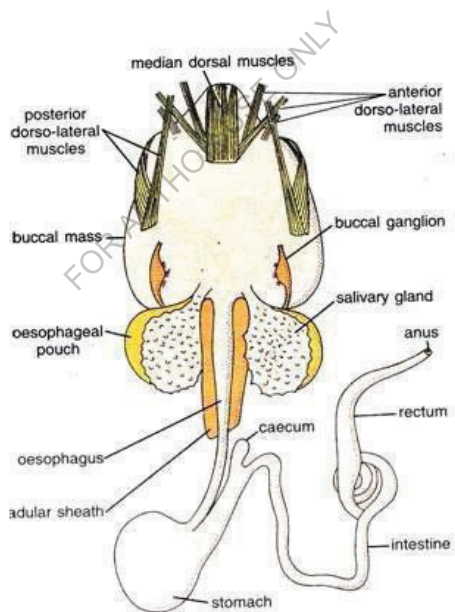


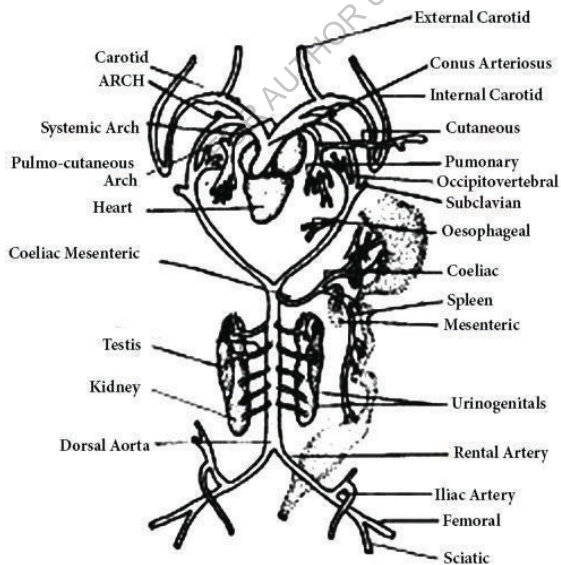
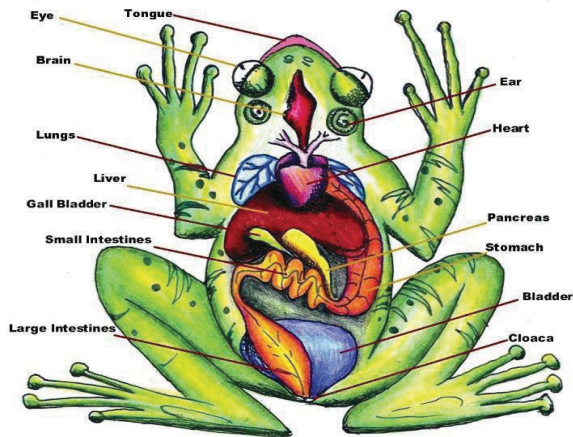
Fig. 1.2 Nervous system of cockroach



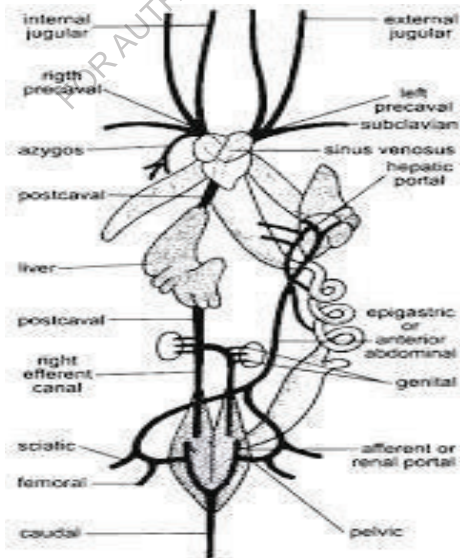
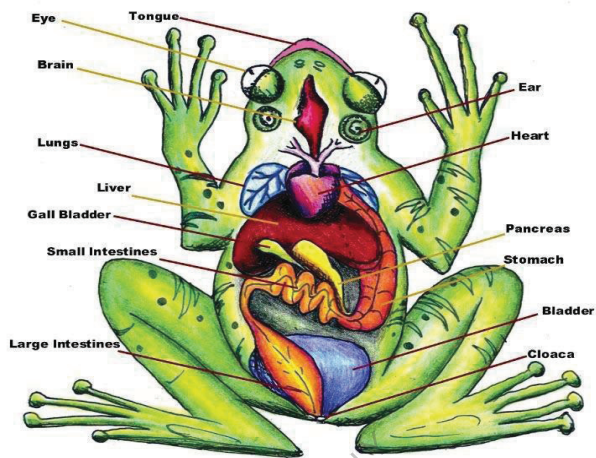
PILA-DIGESTIVE SYSTEM



FROG -ARTERIAL SYSTEM



CALOTUS-VEIN SYSTEM

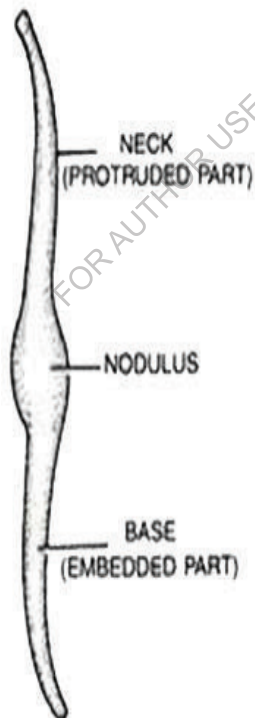


MOUNTING CHARTS

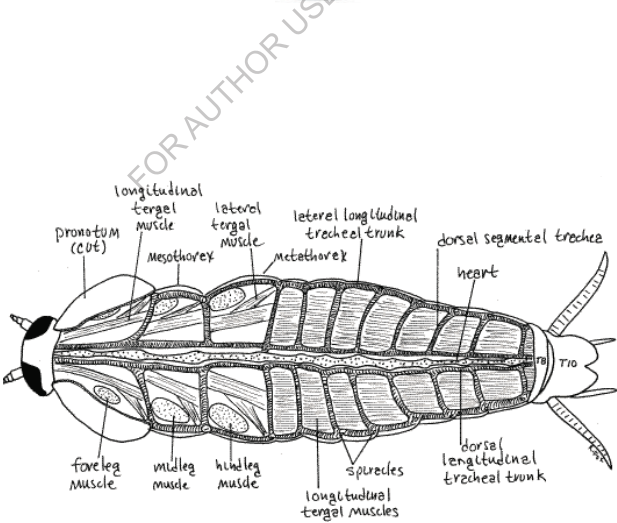
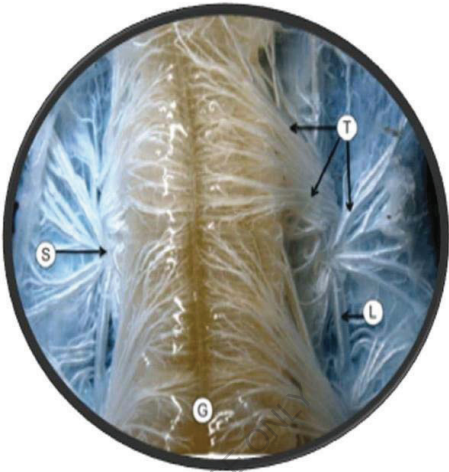
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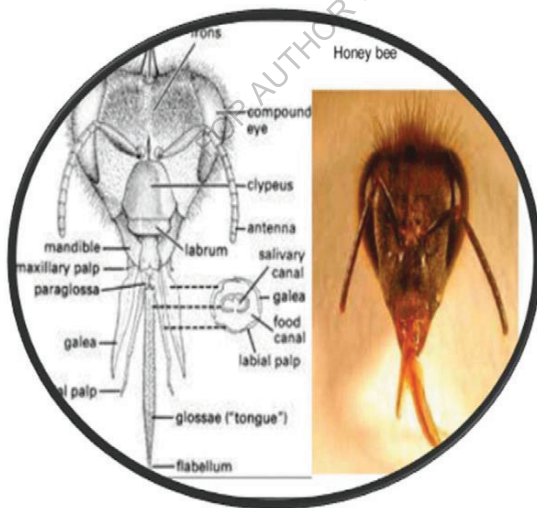
EARTH WORM -BODY SETAE



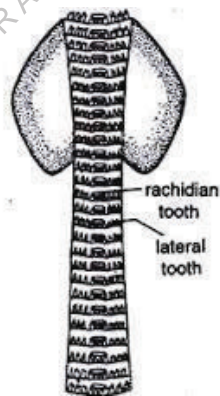
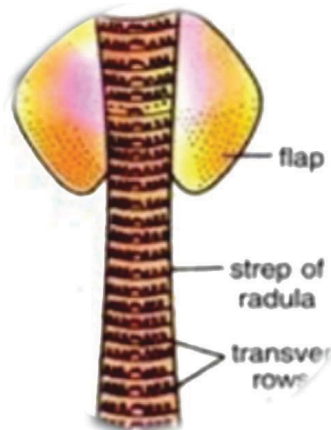
COCKROACH-TRACHEA



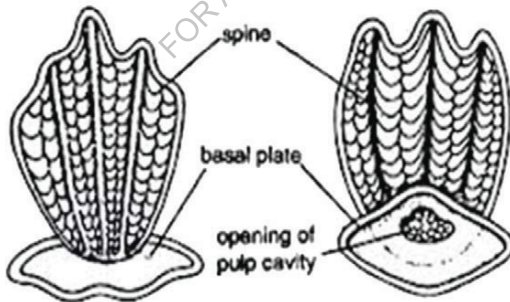
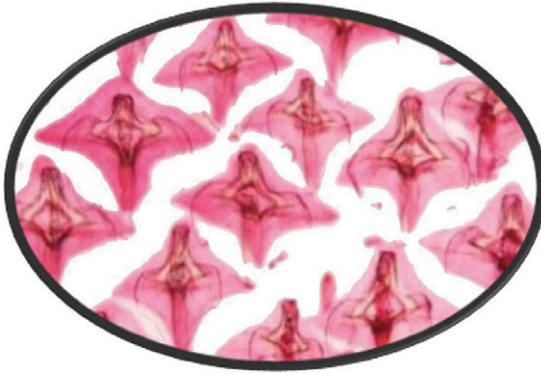
HONEY BEE MOUTH PARTS AND STING



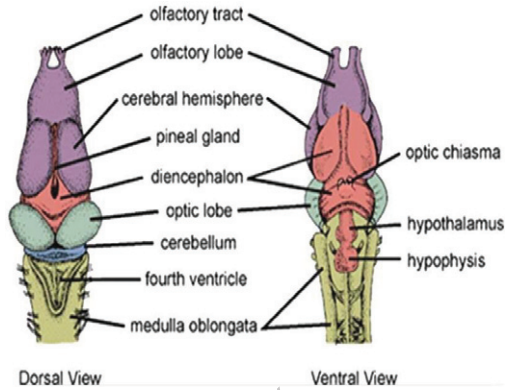
PILA-RADULA



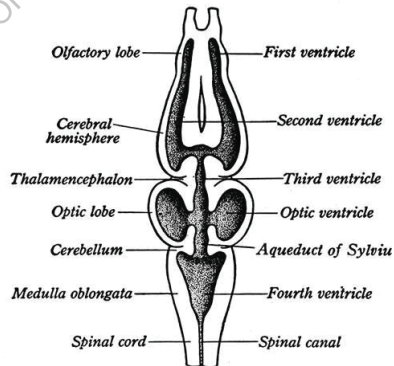
SHARK-PLACOID SCALES



FROG BRAIN



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SPOTTERS

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PROTOZOA

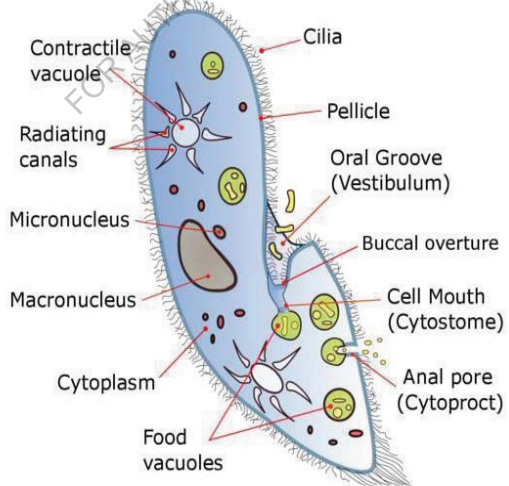
The Protozoa are heterogeneous assemblage of some 50,000 cellular or single- cell organism found worldwide in most habitats.

- Protozoa means „first animal“ the simplest form of animal life. Protozoa are unicellular microorganisms (eukaryotes) that lack cell walls. They are found in marine habitat or soil, in fresh water bodies, symbiotic, some forms are parasites in other organisms. Protozoa depends on nutrition, temperature, pH, and some depends on sunlight.
- Protozoa were defined as unicellular protists with animal-like behavior, such as movement.
- Protozoa were regarded as the partner group of protists to protophyta, which have plant-like behavior, e.g. photosynthesis.
- The term protozoan has become highly problematic due to the introduction of modern ultra-structural, biochemical, and genetic techniques, which have showed that the group does not form a clade as required by modern classifications.
- Modern unicellular clades within Eukaryotes which may be viewed as approximately collectively replacing the class of protozoa include: Excavata, Amoebozoa, Chromalveolata and Rhizaria.

They are restricted to moist or aquatic habitats (i.e., they are obligate aquatic organisms). Many protozoan species are symbionts, some are parasites, and some are predators of feces bacteria and algae

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PARAMECIUM



PARAMECIUM

SCIENTIFIC CLASSIFICATION:

Phylum ...Protozoa

ClassCiliata

Subclass .Euciliata

Order... Holotricha

Family...Paramecidae

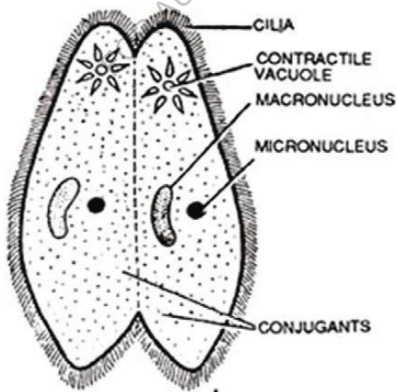
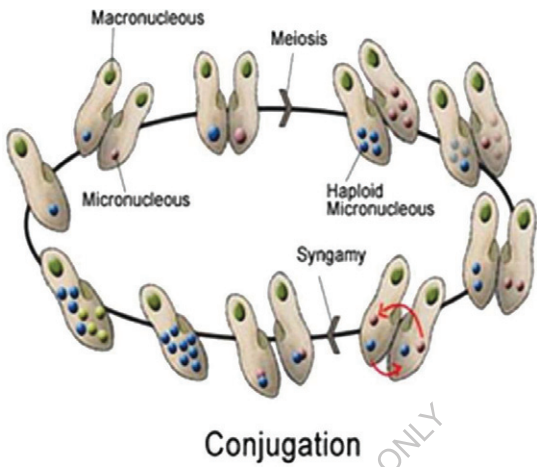
Genus...Paramecium

COMMENTS:

1. Commonly called as slipper animalcule, being microscopic, elongated, slipper- shaped cigar-shaped or spindle shaped.
2. Most familiar and extensively studied protozoan's
3. Pellicle covers the body. It is clear firm and elastic cuticular membrane. Pellicle has series of polygonal or hexagonal depressions for Trichocyst.
4. Paramecia propel themselves by whiplash movements of their cilia, which are arranged in tightly spaced rows around the outside of their body.
5. The beat of each cilium has two phases; a fast "effective stroke," during which the cilium is relatively stiff, followed by a slow "recovery stroke," during which the cilium curls loosely to one side and sweeps forward in a counter-clockwise fashion.
6. The densely arrayed cilia move in a coordinated fashion, with waves of activity moving across the "ciliary carpet," creating an effect sometimes likened to that of the wind blowing across a field of grain.
7. Paramecia live mainly by heterotrophy, feeding on bacteria and other small organisms. A few species are mixotrophs, deriving some nutrients from endosymbiotic algae (chlorella) carried in the cytoplasm of the cell.
8. Osmoregulation is carried out by contractile vacuoles, which actively expel water from the cell to compensate for fluid absorbed by osmosis from its surroundings
9. The number of contractile vacuoles varies from one, to many, depending on species
10. Paramecia reproduce asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis. The cell then divides transversally, and each new cell obtains a copy of the micronucleus and the macronucleus.

IDENTIFICATION: - Since the animal contains slipper- shaped body and two contractile vacuoles while are star – shaped and has all above features hence it is Paramecium.

PARAMECIUM-CONJUGATION

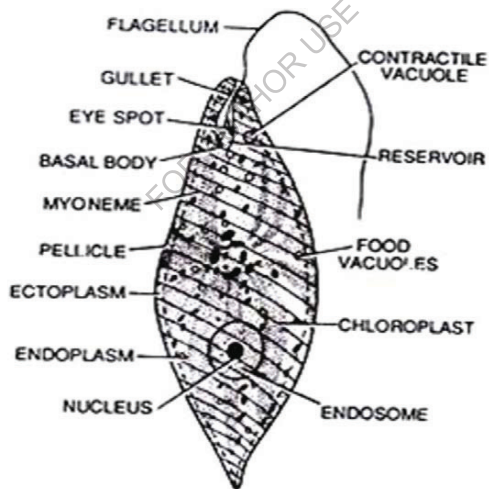
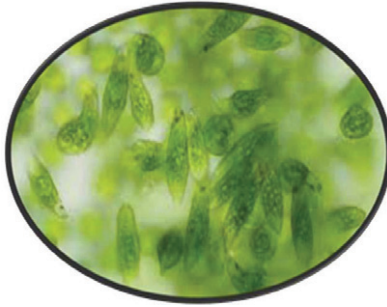


PARAMECIUM-CONJUGATION

SEXUAL REPRODUCTION:

- As mentioned already, paramecium occasionally reproduces by sexual means particularly when the cell is exposed to stressful conditions.
- For this process to take place, two paramecium cells should come together. When this happens, the cells align side by side and remain attached at the oral grooves to each other.
- The micronucleus in each of the fused cells divides by meiosis, leading to formation of four haploid nuclei. Out of these, *three* are aborted and only one survives in each cell.
- The surviving micronucleus again divides mitotically and forms two nuclei.
- The two paramecium cells then exchange one haploid micronucleus and separate.
- It is the meiosis division that leads to exchange of genetic material.
- The new micronucleus fuses with the old to make a diploid micronucleus which then divide into eight tiny micro- nuclei.
- Then, the original macro nucleus slowly disintegrates, and the four micro nucleus expands to form four macro nucleus.
- Later, three out of four micronucleus and macro nucleus separate, leaving the final daughter nucleus with one micronucleus and macro nucleus.
- This sexual reproduction in paramecium is known as conjugation.

EUGLENA



EUGLENA

SCIENTIFIC CLASSIFICATION:

Phylum ...Protozoa

Subphylum....Plasmoda

Class.... Mastigophora

Order... Euglenoidina

Genus... Euglena

COMMENTS:

- Body of animal is simple fuse form, spindle shaped plump like red or green colour.
- Euglena measure 50 to 100 microns in length.
- Outer covering is called as pellicle which are marked by an spiral striations called as myonemes
- Anterior end has funneled shaped cytosome which leads into a flask –shaped cytopharynx.
- On one side is a red mass of hematochrome called stigmata which is photosensitive
- Endoplasm contains nucleus, chloroplast and other ultra-structural organelles
- Nutrition holophytic or saprophytic and reproduction by longitudinal division or encystment.
- Euglena does not take solid food but lives entirely by autotrophic and saprozoic nutrition.
- It Is unique animal with floral mode of nutrition and funnel mode of life and reproduction?
- *Euglena viridis*, *E. rubra*, *E. sanguine* and *E. fusiformis* are common species they respond to various stimuli such as light heat etc.
- Euglena is unique animal with floral mode of nutrition and faunal mode of reproduction.

IDENTIFICATION:

- Since the animals have chloroplast and all above features, hence it is Euglena.

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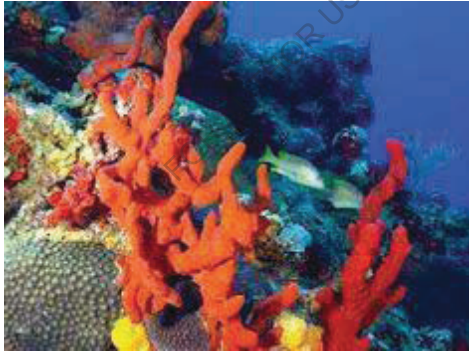
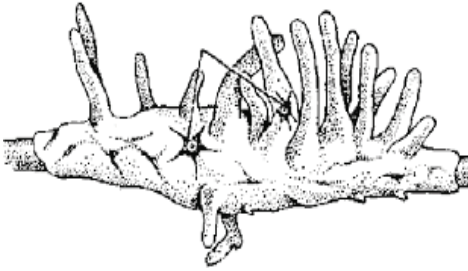
PORIFERA

INTRODUCTION:

Animals belonging to phylum Porifera are commonly called Sponges as these bear pores i.e. they are porous and they are sessile animals which remain attached to rocks, shells or other substrata, growing like plants.

- Sponges are marine organisms distributed from Arctic to Tropic seas except those belonging to family Spongillidae, distributed in freshwater. Body shape is cylinder or vase like, asymmetrical or radially symmetrical.
- These are multicellular organisms having diploblastic body consisting of outer layer of ectoderm, inner endoderm and an intermediate mesenchyme. Spongocoel is the internal space of the body which is either hollow or permeated by numerous canals lined with choanocytes (collared flagellated cells).
- Canal system is present which helps in nutrition, respiration, excretion and reproduction in sponges. Mouth is absent and digestion is intracellular. Skeleton consists of spicules (calcareous or siliceous) or fibres (spongin).
- Asexual reproduction (by buds and gemmules) as well as sexual reproduction (by ova and sperms) takes place in sponges. Fertilization is internal, holoblastic cleavage, indirect development including free-swimming ciliated larval stage called Amphiblastula or Parenchymula.
- Sponges exhibit great power of regeneration.

SIMPLE-SPONGE



SIMPLE-SPONGE

SCIENTIFIC CLASSIFICATION:

PhylumPorifera

ClassCalcarea

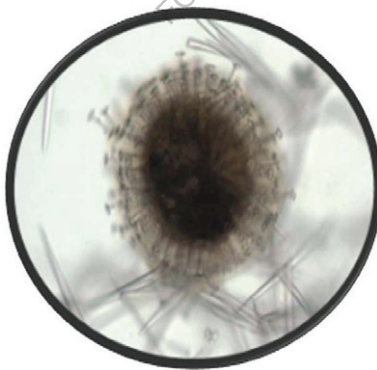
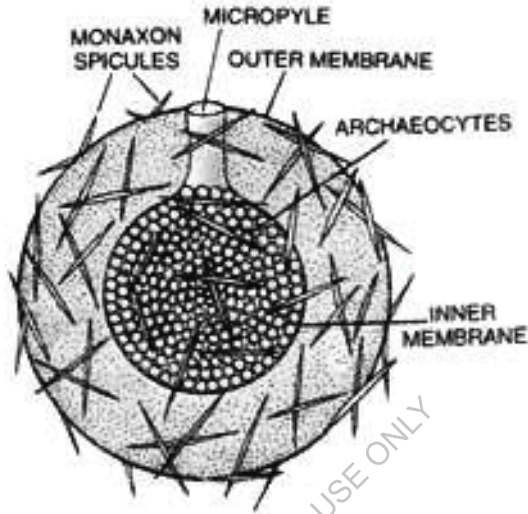
Order.....Homocoela

Genus.....Leucosolenia

COMMENTS:

- Porifera include the sponges which are most primitive of multicellular animals, they are sessile, plant-like animals, they are fixed to some submerged solid rock or shell and are incapable of any movement.
- The porifera are exclusively marine except for a single family of freshwater species.
- Their shape may be cylindrical, branching, vase-like or globular, some are dull in colour but most are brightly coloured, they have red, orange, purple, green or yellow colour.
- The body is perforated by pores and canals but there are no organs, such as mouth or nervous system.
- Though sponges are multicellular animals their cells do not form organized tissues. They usually have an endoskeleton of separate spicules.
- Digestion takes place within the cells. Because of their endoskeleton and obnoxious ferments they are generally not taken by animals.
- Sponges are cultivated for commercial purposes. Approximately 10,000 species of sponges are known at present, and the phylum is divided into three classes, viz., Calcarea or Calcispongiae, Hexactinellida or Hyalospongiae, and Demospongiae and about twelve orders chiefly on the type of skeleton.

SPONGE - GEMMULES



SPONGE - GEMMULES

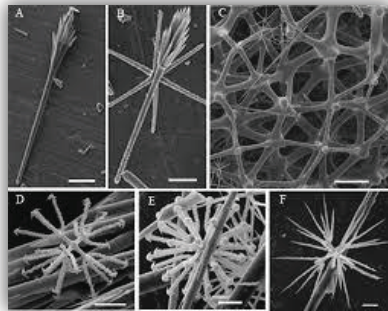
COMMENTS:

- Gemmule are asexual reproductive bodies forming a part of regular life-cycle.
- Gemmule or endogenous buds are diagnostic of Porifera and especially of fresh-water and a few marine sponges.
- Gemmulation or endogenous budding is a peculiar mode of reproduction under unfavorable conditions such as excessive cold or draught.
- Gemmule contain outer and inner membrane.
- Gemmule is rounded structure formed by the aggregation of archaeocytes into groups accompanied by trichocytes which are impregnated with food particles of glycoproteins or lipoproteins.
- Scleroblasts secrete the amphidiscs spicules, which form a row in columnar layer between outer and inner membrane.
- Gemmule is resistant to external factors such as freezing and drying. Gemmule of fresh water sponge can be kept for 2 years.
- They hatch at a temperature of 13-21 °C in about 3 days. After hatching, a gemmule gives rise to your sponge.
- A full grown gemmule is usually pierced by opening on one side, called a micropyle.

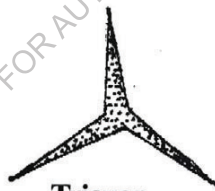
IDENTIFICATION:-

Since the material has micropyle in mature and amphidiscs spicules in immature Gemmule and has above all features, hence it is Gemmule whole mount

SPONGE – SPICULES



Monaxon



Triaxon



Tetraaxon

Spicules of Olynthus.

SPONGE – SPICULES

COMMENTS:-

1. Sponge body wall is supported by various minute, crystalline and bodies called as spicules, which are secreted by special mesenchymal cells as Scleroblasts.
2. Spicules provide taxonomic character and are classified according to the axes and rays, spoken of as axon, actine or actinal respectively.
3. Spicules are of two types. (i) Megascleres or supporting skeleton, (ii) Microscleres small and non- supporting. Kinds of Megascleres are as follows;
4. Monaxon consists of a single axis, straight or curved. They may be styles, rhabds and tylots.
5. Tetraxon consists of four rays. It also includes triradiate or triactinal spicules.
6. Polyaxon having several equal rays. Amphidiscs spicules are found in fresh water sponges. In this type, the rhabdom contains disks at both ends. The arrangement of different types of spicules could be seen in Sycon.
7. Microscleres are found throughout the mesenchyme and include spires and asters.
8. Spicules support and protect the body. They are helpful in identification,
9. Classification and metabolism.

IDENTIFICATION:-

The clear transparent Monaxon or Triaxon spicules indicate Spicule of sponges.

COELENTERATA

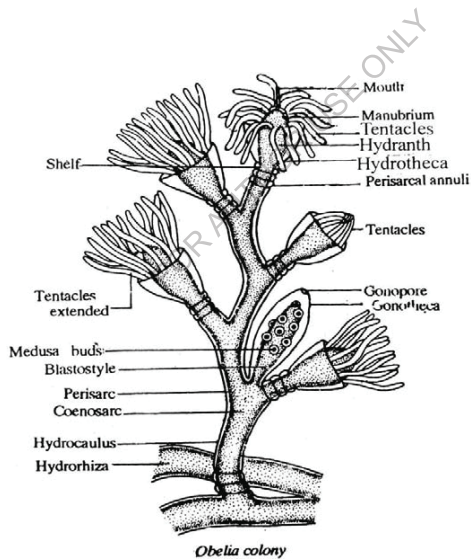
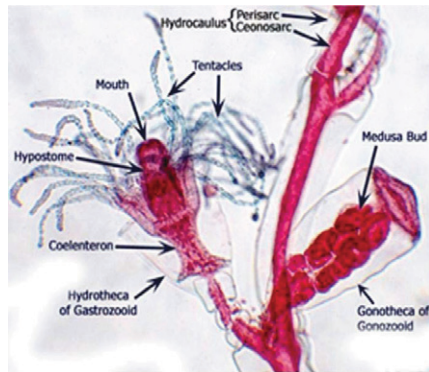
INTRODUCTION:

- These are the Multicellular organisms with tissue grade of organization and are aquatic, mostly marine except a few freshwater forms like Hydra.
- They are sedentary or free swimming, solitary or colonial and show the radial or biradial symmetry with central gastro vascular cavity opening to exterior by mouth.
- Diploblastic body consisting of outer layer of ectoderm, inner endoderm and an intermediate non cellular gelatinous mesogloea.
- Acoelomate i.e. coelom is absent. Mouth is encircled by tentacles bearing nematocysts (stinging cells) meant for capturing food, adhesion, offence and defense.

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OBELIA COLONY



OBELIA COLONY

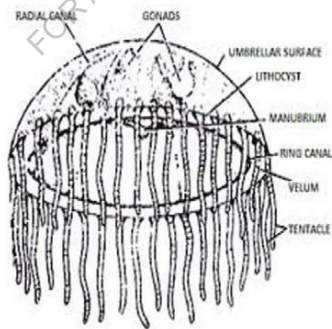
SCIENTIFIC CLASSIFICATION:

- Phylum.....Coelentrata
- Class.....Hydrozoa
- Order.....Hydroidea
- Genus.....Obelia

COMMENTS:-

1. Obelia is commonly called as sea-fur. It is sedentary marine colonial form attached on the surface of sea weeds, molluscan shells, rocks and wooden piles in shallow waters. It is widely distributed throughout the world.
2. It is a trimorphic colony present in the form of filamentous sea weed measuring several centimeters in height.
3. The basal horizontal portion is known as hydrorhiza or stolon. It helps in attachment with the substratum. It gives up vertical branches called hydrocaulus.
4. Hydrorhiza and hydrocaulus are covered by chitinous perisarc which encloses soft inner coenosarc.
5. Coenosarc is the living, hollow cellular tube made up of ectoderm, endoderm and mesoglea.
6. The trimorphic colony of Obelia posses three types of zooids polyp (hydranths), gonangium (blastostyle) and medusa (sexual zooid).
7. Polyp is the nutritive zooid. It is a bell shaped cup made up of lower hydrotheca and upper hypostome. Hypostome carries a ciclet of numerous tentacles provided with nematocysts.
8. Blastostyle is the club shaped reproductive zooid enclosed within a covering called gonotheca. It gives rise to buds which develop into medusa.
9. Medusa is the free swimming reproductive zooid consisting of upper exumbrellar (convex) and lowers sub- umbrellar (concave) sides. It is provided with marginal tentacles, four radial canals bearing gonads, a ring canal and a central hanging manubrium on the concave side.
10. It reproduces by asexual as well as sexual methods. Life history of Obelia exhibits alternation of generation.

MEDUSA OF OBELIA

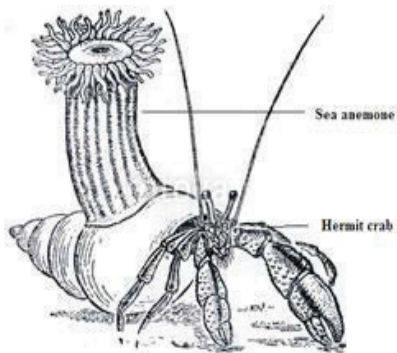


MEDUSA OF OBELIA

COMMENTS:-

1. Mature medusa is free-swimming.
2. Adult medusa is not covered in hydrotheca or in any other protective covering.
3. The manubrium is elongated, tabular comparatively smaller and hangs downwards freely.
4. Mouth is rectangular situated at the tip of manubrium surrounded by 4 oral lobes.
5. The tentacles are 16 in young condition but many in adult stage. The tentacles are situated around the margin of the bell.
6. Medusae are complicated in structure.
7. The bases of tentacles are swollen.
8. The velum is present around the margin of umbrella which is poorly developed.
9. The marginal sense organs, statocysts are present at the bases of 8 radial tentacles.
10. The center on or coelenterons is represented by the stomach, having 4 radial canals and a circular canal.
11. Nervous system is well developed with a double ring of irregular nerve cells.
12. The muscle process of epithelia muscular cells are well developed and constitute a well-developed musculature.
13. The mesoglea is well developed.

SEA ANEMONE



SEA ANEMONE

COMMENTS:

1. It is a common sea- anemone. It lives in shallow sea-water attached to some substratum.
2. The body is cylindrical.
3. The centrally situated mouth is encircled by many circlets of tentacle. The tentacles are with nematocyst.
4. Internally it contains 2 siphon glyphs and 6 pairs of mesenteries which divide the coelenterons in cavities.
5. It exhibits the phenomenon of commensalism.
6. Commensalism is the association between two organisms for mutual benefits both adamasia and hermit crab.

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HELMINTHES

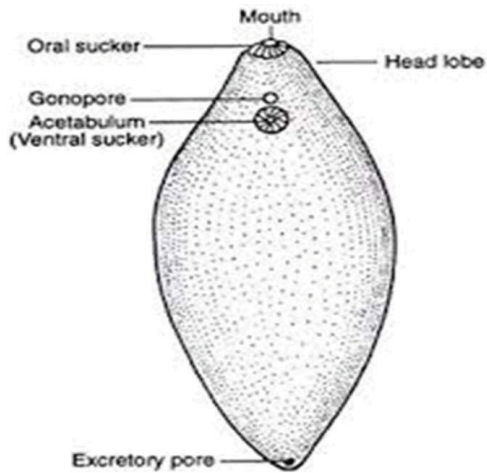
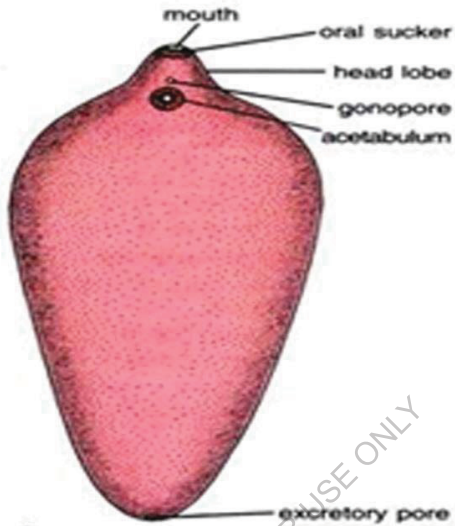
INTRODUCTION:

- Platyhelminths are bilaterally symmetrical and dorso-ventrally flattened.
- Body wall is triploblastic (form 3 germ layers- epidermis, mesodermis and endodermis) below the endodermis longitudinal.
- Circular and oblique muscles are present.
- Acoelomate (true coelom is absent) and unsegmented except class Cestoda.
- They are free living, parasitic or commensals. Adhesive organs like hooks, suckers and spines are present with adhesive secretions.

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LIVER FLUKE



LIVER FLUKE

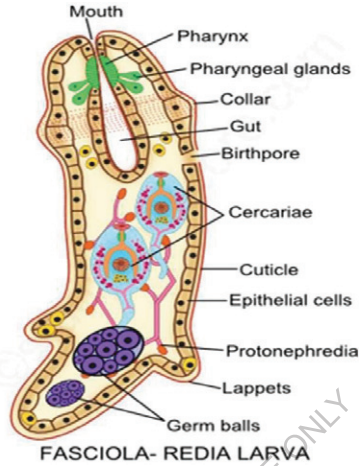
COMMENTS:

1. Fasciola hepatica is found in the bile duct
2. Body is leaf-like, dorso in breadth.
3. Anterior end is produced into a conical projection is known as cephalic cone.
4. Posterior end is large more rounded in front than behind.
5. Mouth is situated ventrally at anterior end and s
6. Digestive system comprised of pharynx
7. They are Hermaphrodite.
8. Excretory pore is found at the posterior extremity.
9. Eggs pass to the exterior through median genital pore situated between oral sucker.
10. Life Cycle comprises of an intermediate host
11. They causes a disease known as liver rot.

IDENTIFYING CHARACTERS:

- It has cephalic cone, acetabulum, anterior end is broad and rounded while posterior end is pointed.

REDIA LARVA



REDIA LARVA

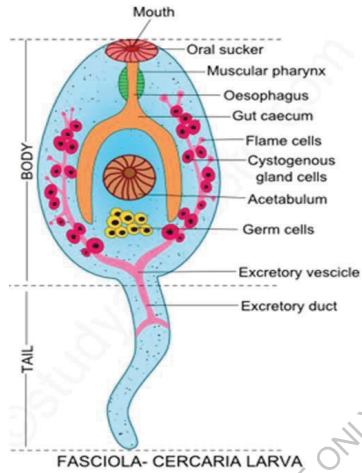
COMMENTS:-

Redia Larva of Fasciola

1. Redia Larva develops when the germ cells of the sporocyst multiplies
2. Each redia is an elongated and cylindrical structure about 1.3 mm to 1.6 mm long.
3. Anterior end bears the mouth, muscular swelling or collar (which helps in locomotion) and a permanent birth pore.
4. Two stumpy processes known as lappets or procruscula ventro posterior end which helpful in locomotion.
5. The space between the body wall and intestine contains few germ cells. It gives rise to second generation the daughter rediae.
6. Daughter redia and cercaria come out from mother redia through birth pore.

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CERCARIA LARVA OF FASCIOLA HEPATICA



CERCARIA LARVA OF FASCIOLA HEPATICA

COMMENTS:-

1. Each redia produces produce 14-20 cercaria larva
2. After leaving the body of redia they enter in the digestive glands of snails.
3. Body of cercaria is and it is free swimming.
4. Body wall consists of cuticles, muscles and mesenchyme.
5. It has higher grade of organization and close resemblance with the adult fluke.
6. It has two suckers (i) anterior oral sucker surrounding the mouth and (ii) ventral sucker situated in the middle of the body.
7. Body and tail are covered with backwardly directed spines.
8. Digestive system comprises mouth, oral sucker, muscular pharynx, esophagus and inverted U shaped (forked) fork of intestine. 9. Flame cells are present as excretory organ and opens into a pair of excretory tubule which unite in front of tail to form an excretory vesicle or bladder.
10. Body space is filled with parenchyma form the cyst of the next larva. 11. Rudimentary reproductive organs (Genital rudiments) are also present.
12. Cercaria larva has a very active life. After 2 or 3 days of active life its loses its tail and undergoes encystment to become *Meta cercaria* larva.
13. The encysted larva (*Meta cercaria*) is finally swallowed by sheep

ASCHELMINTHES

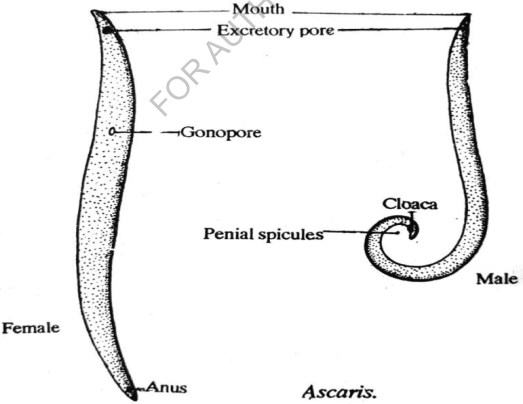
INTRODUCTION:

- Phylum Aschelminthes (Round worms) inhabit aquatic and terrestrial environment and are well known parasites.
- They are bilaterally symmetrical and unsegment worms. Triploblastic and pseudocoelomate animals with organ system grade of body organization.
- Body size mostly small, some microscopic, while others few millimeters to a meters or even more in length.
- Body mostly cylindrical or flattened, body wall with cuticle and cilia absent.
- The digestive system includes straight and complete alimentary canal with mouth and anus, muscular and highly specialized pharynx.

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ASCARIS –MALE AND FEMALE



ASCARIS MALE AND FEMALE

SCIENTIFIC CLASSIFICATION:

Phylum: Aschelminthes

Class : Nematoda

Order : Ascaridida

Family: Ascarididae

Genus : *Ascaris*

Species : *lumbricoides*

COMMENTS:-

1. A roundworm called *Ascaris lumbricoides* infects a person's small intestine and acts as an endoparasite, getting nutrients from its host's intestinal tract.
2. It is cosmopolitan in distribution.
3. It shows sexual dimorphism with separate male and female individuals. Males measure 15-30 cm in length and female 20-35 cm.
4. The tail end of male is ventrally curved containing cloacae aperture, through which two spicules project.
5. The tail end of the female is straight. Anterior and end exhibits same structures in both male and female.
6. Mouth is situated at anterior end. It is guarded by three lips.
7. Anaerobic respiration is noticed.
8. Excretory pore lies at the distance of 2 mm. From anterior end. No intermediate host is required.
9. Life cycle includes a rhabdiform larva and an extra intestinal migration.

Prevention:

- People should also wash any utensils and cooking surfaces after each use.
- Filter and boil water before drinking it.
- Wash hands with soap and hot water before handling food and eating.
- Avoid common bathing areas, especially if they appear unclean.
- Peel and cook any vegetables or fruit grown in unsanitary conditions.
- Ensure children clean their hands with soap and water after playing outside.

Treatment:

The worm will treat most cases of ascariasis with antiparasitic drugs like Albendazole (Albenza), Ivermectin (Stromectol) and Mebendazole.

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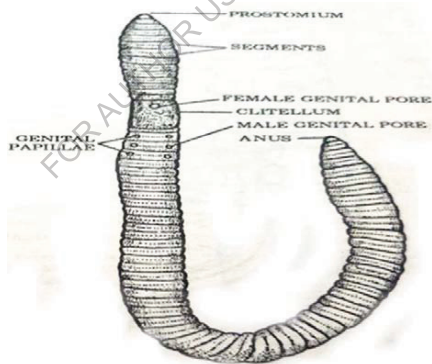
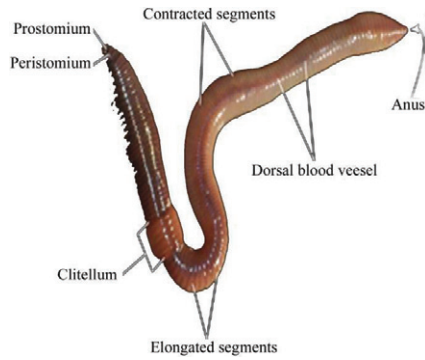
ANNELIDA

INTRODUCTION

- Annelida (L., annulus = little rings) known as ringed worms or segmented worms are mostly aquatic, marine or freshwater, some burrowing or tubicolous, terrestrial, free living or sedentary forms.
- Body elongated, Triploblastic, bilaterally symmetrical, metamerically segmented, truly coelomate and vermiform. Body has more than two cell layers, tissues and organs.
- Epidermis of a single layer of columnar epithelial cells, covered externally by a thin cuticle not made of chitin.
- Body wall is contractile, consists of an outer epidermis, longitudinal and circular muscles. Appendages when present are un jointed.
- Nervous system consists of a anterior nerve ring and segmental ganglia connected by ventral nerve cord.
- Digestive system straight and complete.
- Blood vascular system is closed type. Respiratory pigment is hemoglobin or erythrocrurin dissolved in blood plasma.
- Respiration through general body surface and by gills.
- Locomotory organs are Segmentally arranged paired setae.
- Excretory system consists of paired Nephridia that excrete nitrogenous waste.
- Sexes may be separate or united. Development is direct and indirect. Direct development in monoecious and indirect development in Dioecious forms.
- Trochophore larval stage occurs in case of indirect development.

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EARTH WORM



EARTH WORM

SCIENTIFIC CLASSIFICATION:

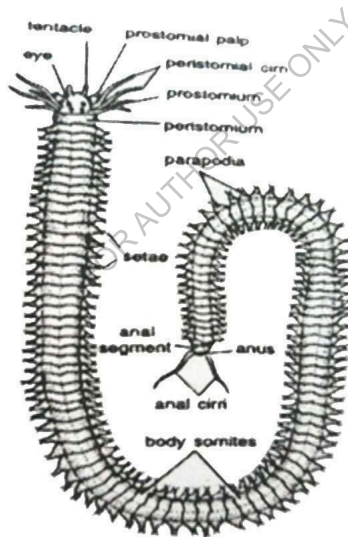
- Phylum : Annelida
- Class : Oligochaeta
- Order : Neooligochaeta
- Genus : Pheretima

COMMENTS:

1. Pheretima is commonly called earthworm.
2. Body is bilaterally symmetrical, narrow, long, elongated and cylindrical measuring up to 150 mm in length.
3. Anterior end is tapering and posterior end is blunt.
4. Body is divided into 100-120 ring-like segments by a distinct series of annular grooves.
5. Each segment is provided with setae arranged in a ring with each setae arising from a setigerous sac of the skin. The setae help in locomotion by holding the earth.
6. Mouth is crescent aperture situated just below the prostomium.
7. The clitellum is a circular band of glandular tissue which completely surrounds the segments from 14th to 16th segment.
8. Hermaphrodite.
9. A pair of male genital pore is situated ventrally in the eighteenth segment while female genital pore are situated at the ventral surface of fourteenth segment.
10. Anus is situated at the terminal end of the last body segment called anal segment.
11. Reproduction is usually sexual.
12. Development takes place in cocoons.
13. Earthworm is used as bait in fishing and as food by many uncivilized people. It also has use in medicines, education, experiments and in agriculture as producer of organic fertilizer

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NEREIS



NEREIS

SCIENTIFIC CLASSIFICATION:

Phylum :Annelida

Class : Polychaeta

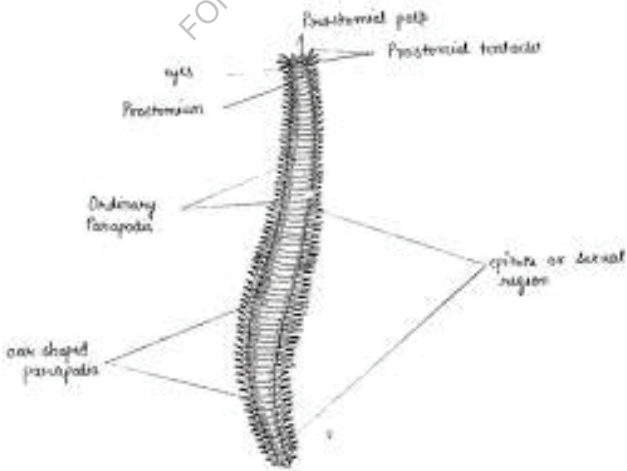
Order : Errantia

Genus : Nereis

COMMENTS:

1. Nereis is commonly called rag worm.
2. The body is long, slender, elongated, Dorso-ventrally flattened segmented and is divisible into head, trunk and Pygidium.
3. Head consists of two parts, the prostomium and Peristomium. Prostomium bears a pair of tentacles, two pairs of eyes and a pair of short two jointed palps. Peristomium bears four tentacles and a slit-like mouth.
4. Trunk is made up of several segments, each bearing a pair of lateral Parapodia which are Locomotory organs. Setae project beyond the outer margin of each Parapodium.
5. Pygidium or anal segment is without Parapodia but bears a pair of appendages known as anal cirri and a terminal anus.
6. Respiration happens via blood capillary network of Parapodia.
7. Alimentary canal is straight and extends from mouth at the anterior end to the anus at the posterior end.
8. Sexes separate. Fertilization is external.
9. The sexual phase of Nereis is known as Heteronereis.

HETERONEREIS



HETERONEREIS

SCIENTIFIC CLASSIFICATION:

Phylum: Annelida

Class : Polychaeta

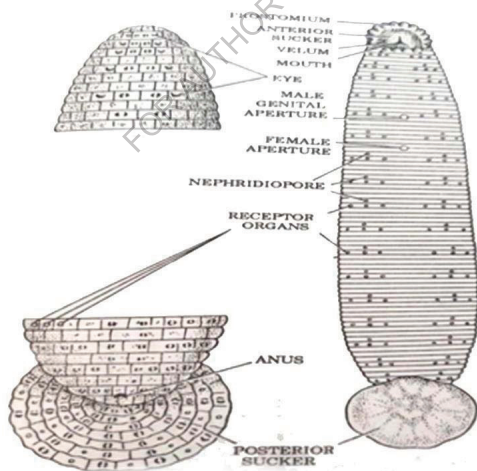
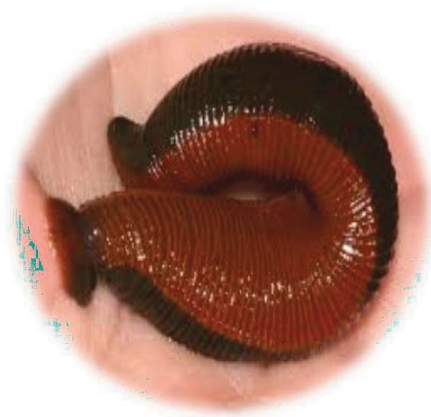
Order : Errantia

Genus : Heteronereis

COMMENTS:

1. Heteronereis is the sexual phase of Nereis.
2. The body of Heteronereis is divisible into an anterior atoke or asexual region and a posterior epitoke or sexual region.
3. It comes out to the water surface and leads an active free swimming life.
4. Eyes become greatly enlarged and highly sensitive to light.
5. The peristomial cirri become longer.
6. Para podia become enlarged develop additional foliaceous outgrowths and setae become oar shaped which help in active swimming.
7. Due to excessive development of gonads, the muscles and alimentary canal are reduced.
8. There is no marked sexual dimorphism in most species.

LEECH (Hirudinaria)



LEECH (Hirudinaria)

SCIENTIFIC CLASSIFICATION:

Phylum..Annelia

Class....Hirudiea

Order...Gnathobdella

Genus : Hirudinaria

COMMENTS:-

1. *Hirudinaria granulosa* is commonly called Indian cattle leech.
2. Body is soft, elongated, vermiform and bilaterally symmetrical with dorsal surface green and ventral surface orange yellow in colour.
3. During extension of the body is Dorso-ventrally flattened while during contraction it is cylindrical.
4. Body of leech is divided into 33 segments. The segments are further divided into annuli or rings.
5. Anterior and posterior suckers are well developed. Anterior sucker is oval bearing tri radiate mouth and is formed by the fusion of prostomium with a few somites of anterior region. The posterior sucker is circular in outline and forms a muscular disc at the posterior end. Both the suckers serve as powerful organs of adhesion and locomotion.
6. Five pairs of eyes are present on the dorsal side.
7. No special respiratory organs. Respiration happens via skin.
8. Locomotion by creeping/crawling on the surface.
9. Alimentary canal is a straight tube extending from mouth to anus. Small aperture or anus is situated middorsally on the 26th segment at the root of the posterior sucker.
10. Excretory system consists of Segmentally arranged seventeen pairs of Nephridia arranged from six to twenty-two segments, opening by nephridiopores on the ventral surface.
11. Hermaphrodite. Male genital aperture is situated mid-ventrally in the second and third annuli of 10th segment and female genital aperture between second and third annuli of 11th segment.
12. Sexual reproduction is common.

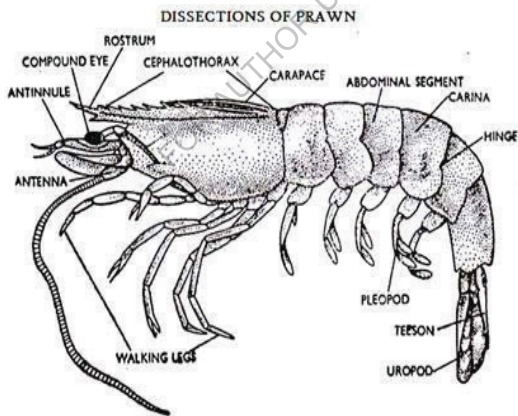
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ARTHROPODA

INTRODUCTION:

- Phylum Arthropoda (Gr., arthros = joint + podos = foot) is the largest phylum and most varied in the animal kingdom. Von Siebold gave the name Arthropoda.
- It includes well over one million described species and many million remain unstudied. Some of the more well-known arthropods include insects, crustaceans, spiders, scorpion and centipede as well as the fossil trilobites.
- Arthropods are mainly terrestrial but marine and freshwater species are also well known.
- Arthropoda varies tremendously in their habitats, life histories, and dietary preferences. Arthropod bodies are divided into segments. However, a number of segments are sometimes fused to form integrated body parts known as tagmata.
- This process of fusion is called tagmosis. The head, thorax, and abdomen are examples of tagmata. Exoskeleton serves as protection and provides places for muscle attachment.
- Arthropods must molt because their exoskeletons don't grow with them. However, in most species some appendages have been modified to form other structures, such as mouthparts, antennae or reproductive organs.
- Arthropod appendages may be either biramous (branched) or uniramous (unbranched). In insects, the anterior portion of the heart is extended into a tube that is called an aorta which directs the blood forward as it goes out into the body cavity

PRAWN (Palaemon)



PRAWN (Palaemon)

SCIENTIFIC CLASSIFICATION:

Phylum: Arthropoda

Class :Crustacea

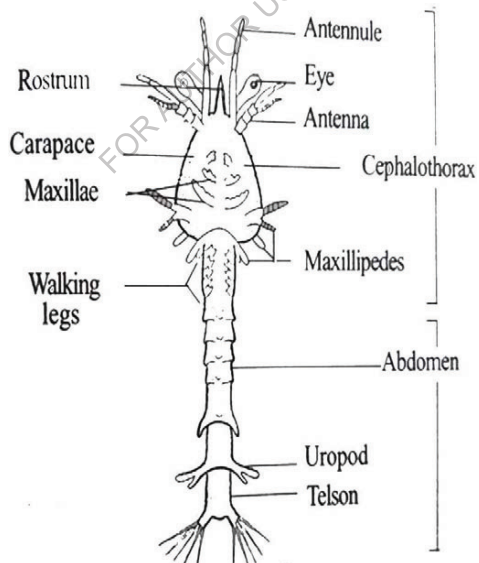
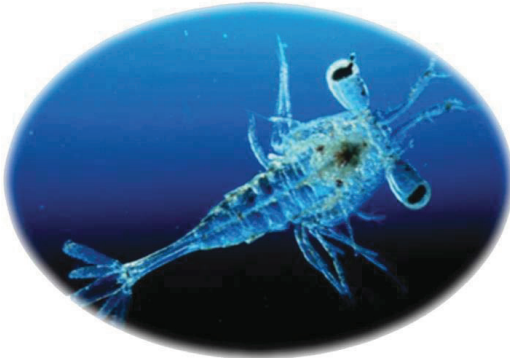
Order : Decapoda

Genus : Palaemon

COMMENTS:-

1. Palaemon is commonly known as prawn.
2. Body is elongated, spindle shaped and bilaterally symmetrical.
3. Palaemon species are of pale-yellow, pale-blue and greenish color with brown tinge or with orange-red patches. Preserved specimens become deep orange-red.
4. The body can be divided into two regions, anterior cephalothorax and posterior abdomen.
5. Cephalothorax is a large, rigid, un jointed, immovable and cylindrical structure. It consists of 13 segments, 5 of the head region, and 8 of the thorax region
6. Abdomen is rounded, jointed and compressed laterally. It consists of 6 movable segments and a terminal conical structure, called telson. Each abdominal segment bears a pair of jointed appendages called pleopods or swimmerets.
7. The prawn uses its walking legs for movement at the water-bed.
8. Respiration happens via gills, epipodites and lining of branchio stegites.
9. Excretion through a pair of antennary or green glands, a pair of lateral ducts, an unpaired renal or Nephroperitoneal sac and the integument.
10. Sexes are separate. Sexual dimorphism is well developed.

ZOEALARVA



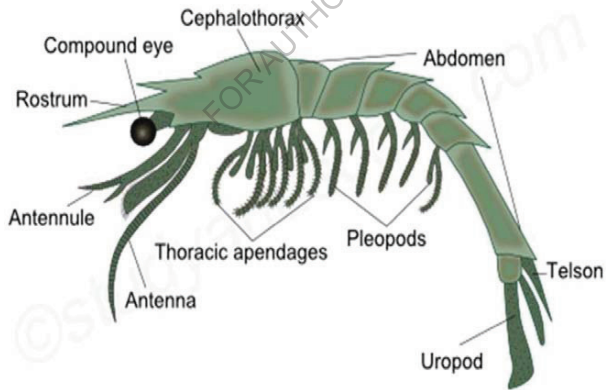
ZOEALARVA

COMMENTS:-

1. Zoea larva is a larval stage of the crustaceans.
2. Body consists of an unsegment cephalothorax and segmented abdomen.
3. Cephalothorax is covered by a carapace and has a pair of compound eyes
4. Head contains one dorsal, one rostrum and two lateral spines.
5. It contains mandibles, maxillae, antennules, antennae and two pairs of well developed maxillipedes.
6. Six pairs of thoracic appendages develop as buds. It uses thoracic appendages for swimming.
7. Abdomen comprises of six segments and bears caudal fork.

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MYSIS LARVA



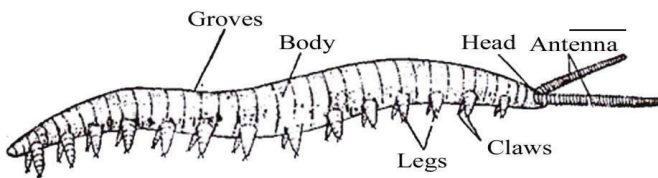
MYSIS LARVA

COMMENTS:

1. Mysis is a stage in Arthropoda such as Peanaeus.
2. The body appears similar to the adult Mysis.
3. Body is divisible into cephalothorax and abdomen.
4. Cephalothorax bears a pair of compound eyes, a rostrum, a pair of antennules and a pair of antennae.
6. It bears seven pairs of biramous thoracic appendages and used for locomotion.
7. Abdomen bears five pairs of biramous pleopods. The last abdominal segment has a telson and a pair of uropods.
8. Mysis larva metamorphoses into the adult prawn.

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PERIPATUS

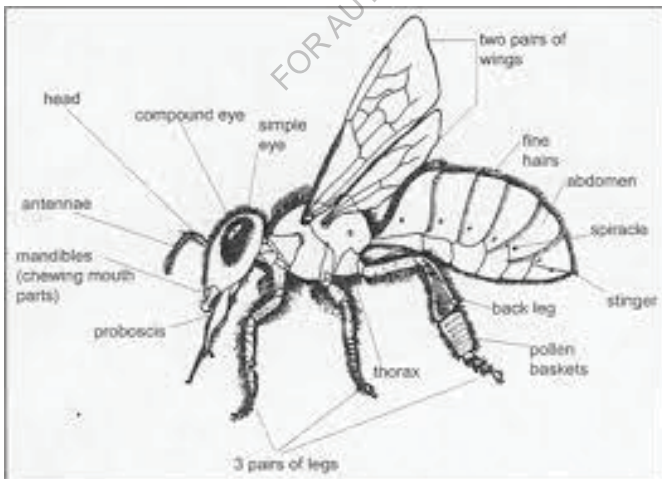
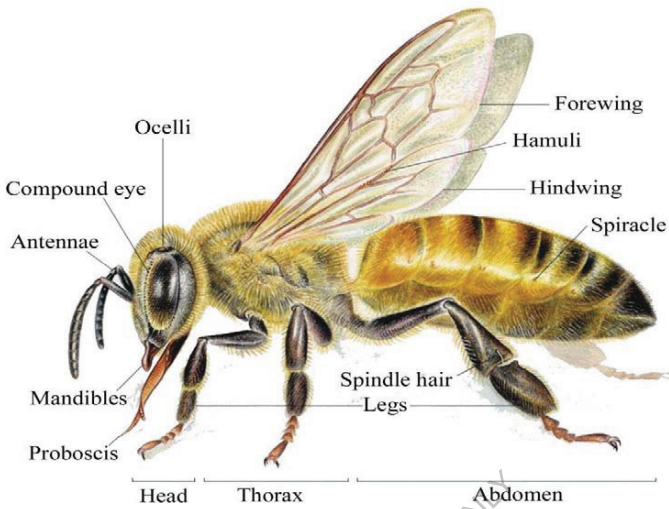


PERIPATUS

COMMENTS:-

1. It is found in the forests of Malaya, Australia, New Zealand, Africa, Mexico, Central America and West Indies.
2. It is shy and nocturnal and lives below bark and stanes or holes in rocks, as it likes dark and damp places.
3. The body is cater-pillar like without any external segmentation.
4. Head with a pair of eyes and pair of segmented antennae.
5. The mouth is mid-ventral in position and has a pair of small jaws.
6. Anus at the posterior end and genital pore a little above the anus in between the pair of legs.
7. A series of short segmented legs with claws on the ventrolateral side of the body.
8. The cylindrical body, structure of the eyes, legs, segmentally arranged paired Nephridia are Annelidan characters. The Peripatus has some of the Annelida characters, on the other hand it has Arthropoda characters like antennae, haemocoel, trachea etc. Thus it represents the living connecting link between the two groups.

HONEY BEE



HONEY BEE

SCIENTIFIC CLASSIFICATION:

Phylum : Arthropoda

Class: Insecta

Order: Hymenoptera

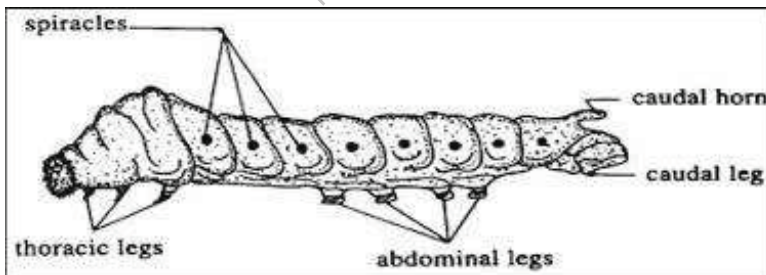
Family: Apidac

Genes: Apis

COMMENTS:-

1. Structure with two Apex pointed below.
2. The mouth parts are attached to the lower part of the head and are of chewing and lapping type.
3. First abdominal segment called propodeum is fused to the thorax. The anterior wings are longer than the posterior pair.
4. There are 3 pairs of legs which are densely covered with hairs, which besides walking, help in collections pollen and are variously modified.
5. Abdomen begins with the second segments 2-7 are clearly visible but segment 8-10 are modified and hidden.
6. The sting is made of 3 pair of gonapophyses one pair of segment 8, and two pairs of 9th segments.

SILK WORM



SILK WORM

SCIENTIFIC CLASSIFICATION:

Phylum : Arthropoda

Class : Insecta

Order : Lepidoptera

Genus : Bombyx

COMMENTS:-

1. Bombyx is commonly called as silk moth.
2. Body consists of head, thorax and abdomen.
3. Head bears a pair of plumed antennae and a pair of compound eyes.
4. Thorax bears three pairs of legs and two pairs of wings which are covered with scales.
5. Hindwing is smaller than forewing.
6. The abdomen is consists of ten segments.
7. Mouthparts are siphoning type.
8. Moths are nocturnal. They come on the light during the rainy season.
9. Habit and habitat: Bombyx is reared for silk. The adults have a short lifespan and don not feed.

DISTRIBUTION:

Bombyx are cosmopolitan in distribution.

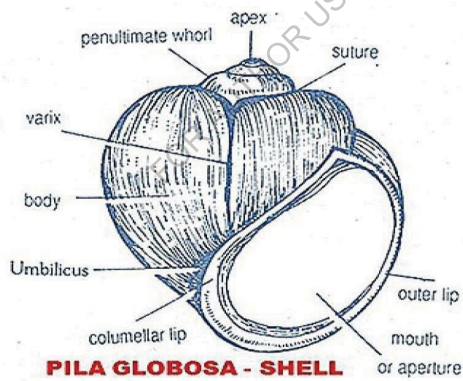
MOLLUSCA

INTRODUCTION:

- *Phylum mollusca* (L., molluscus, soft) includes soft-bodied invertebrate animals such as Octopods, snails, slugs, mussels, clams, oysters, tusk-shells, squids etc.
- The term mollusca were coined by Johnston. It is a diverse and widespread group, with about 112,000 species.
- Molluscs are the largest marine phylum, comprising about 23% of all the named marine organisms.
- Molluscs may be terrestrial or aquatic (freshwater or marine). They are found in all habitats - deserts, forests, lakes, rivers, abysses of sea, coral reefs, underground or even as parasite in the body of other animals.
- They may be found clinging to the rocks, crawling, and swimming, burrowing or even digging. They have diversified feeding habit and can be herbivores, carnivores or omnivores.
- They vary in size from giant squids (nearly 2m long) to little snails, a millimeter long.
- The largest of the molluscan classes is Gastropoda, which is represented by about 35,000 living and some 15,000 fossil species.
- The three most universal features defining molluscs are - a mantle with a significant cavity used for breathing and excretion, the presence of a radula, and the structure of the nervous system.
- The "generalized" mollusc's feeding system consists of a rasping "tongue", the radula, and a complex digestive system. Although molluscs are coelomates, the coelom tends to be small, and the main body cavity is a hemocoel through which blood circulates; their circulatory systems are mainly open.
- The generalized mollusc has two paired nerve cords, or three in bivalves. Most molluscs have eyes, and all have sensors to detect chemicals, vibrations, and touch. The simplest type of molluscan reproductive system relies on external fertilization, but more complex variations occur. All produce eggs, from which may emerge trochophore larvae, more complex veliger larvae, or miniature adults.
- Molluscan shells have always been economically important, having served as money in early days. They have been used in jewellery and buttons. Octopus, Scallops, oysters and squids are important food items. The scientific study of molluscs is called malacology.

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PILA (PILA GLOBOSA)



PILA (PILA GLOBOSA)

SCIENTIFIC CLASSIFICATION:

Phylum : Mollusca

Class : Gastropoda

Order : Mesogastropoda

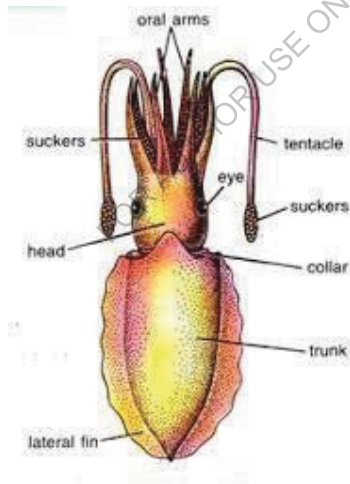
Genus : *Pila*

Species : *globosa*

COMMENTS:-

1. *Pila globosa* is commonly called apple snail.
2. Shell is univalve and coiled around a central axis in a right handed spiral.
3. Operculum is well developed and closes the aperture or the mouth of the shell.
4. The body is soft and is enclosed in a shell. It consists of head, foot and visceral mass.
5. Head is the anterior fleshy part of the body overhanging the foot. It bears mouth, two pairs of contractile tentacles and a pair of eyes.
6. Foot is the Locomotory organ of *Pila* and lies below the head.
7. All visceral organs are contained in this lump like structure that lies above the head-foot complex.
8. Skin of the visceral mass forms a thin and delicate covering called the mantle.
9. Respiratory organs consist of a single ctenidium or gill, a pulmonary sac or lung and a pair of Nuchal lobes. Aquatic respiration by ctenidium and aerial respiration by pulmonary chamber.
10. Sexes are separate but without sexual dimorphism.
11. It is found in large numbers in those areas which are rich in aquatic plants like *Vallisneria* and *Pistia*.

SEPIA



SEPIA

SCIENTIFIC CLASSIFICATION:

Phylum: Mollusca

Class : Cephalopoda

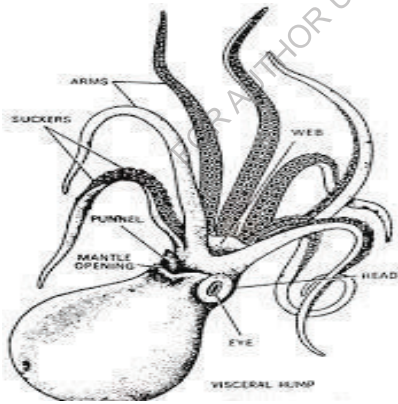
Order : Decapoda

Genus : *Sepia*

COMMENTS:

1. Sepia is commonly called cuttle fish which is a misnomer because it is actually a mollusc and not a fish.
2. Body is soft, un segmented and bilaterally symmetrical. It is divisible into a large head, a small neck or collar and trunk.
3. The head is well developed, bears a pair of eyes and 10 oral arms which are a modification of foot. 8 oral arms are smaller while two are longer and are called tentacles.
4. Collar is constricted and connects the head with the trunk.
5. The trunk extends into flaps called lateral fins on either side that help the animal in swimming.
6. It is covered by a thick muscular mantle, enclosing a large mantle cavity, on the ventral side which contains viscera.
7. It has a life expectancy of 1-2 years.
8. The Sexes are separate and reproduction is sexual.
9. Ornament makers use cuttle bones as moulds for casting small objects.
10. Sepia is a marine animal that resides shallow to mid water and is often associated with coral reefs. It is carnivorous, feeds on crustaceans, small fishes and other animals.

OCTOPUS



OCTOPUS

SCIENTIFIC CLASSIFICATION:

Phylum : Mollusca

Class : Cephalopoda

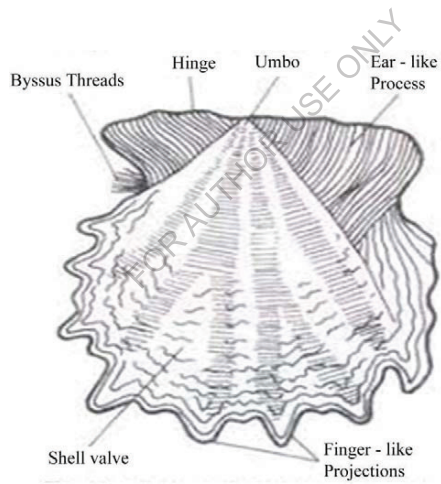
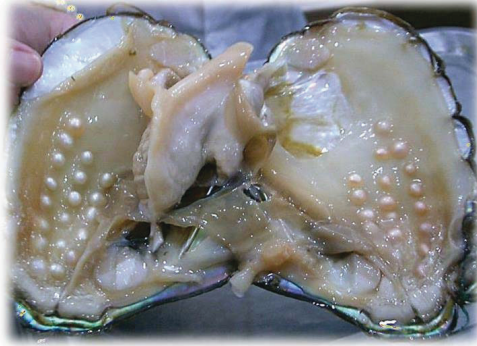
Order : Octopoda

Genus : *Octopus*

COMMENTS:

1. An Octopus is commonly called devil fish.
2. The body is un segmented, symmetrical and soft bodied animals.
3. The head bears a pair of eyes. The mouth is surrounded by eight elongated equal arms usually bearing suckers.
4. Shell is absent.
5. It moves around by crawling or swimming.
6. For defense it ejects ink from the ink-gland into the surrounding water, producing a smoky cloud.
7. In males one of the arms, called hectocotyli zed arm, bears a spoon shaped organ at its end. The arm is used to caress the female and deposit spermatophores beneath its mantle.
8. Sexes are separate.
9. Development is direct.
10. It feed upon crabs, bivalves and fishes etc.
11. Octopus is a nocturnal marine creature inhabits many diverse regions of the ocean, including coral reefs, pelagic waters and the ocean floor.
12. Octopus is cosmopolitan in distribution. It is general.

PEARL OYSTER (*Pinctada vulgaris*)



PEARL OYSTER (*Pinctada vulgaris*)

COMMENTS:

1. Oyster is a shell fish. Hence oyster is shell fish culture. It is bivalve mollusk.
2. It is found attached to the rocks. One shell is cemented to the rock and the other shell acts as a lid.
3. Edible oyster is a Marine Mollusc. It is a filter feeder.
4. Oyster stores glycogen rather than lipids. The stored glycogen is responsible for its unique taste.
5. Spawning takes place during summer season when the temperature is high. In warm water, oyster spawn throughout the year.
6. Oysters are able to change sex, starting as males and later changing to female as they grow. The sex change occurs after the completion of spawning.
7. The eggs and sperm are released into the water fertilization is external.
8. The fertilized egg develops into a larva. The larva settles on some solid objects and develops into young oysters called spat.

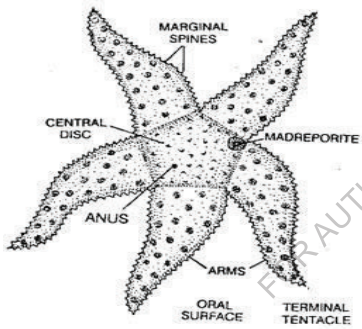
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ECHINODERMATA

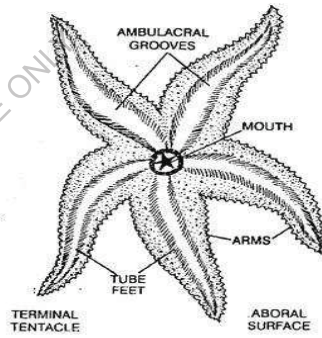
INTRODUCTION

- Phylum Echinodermata (Gr., Echinus=spiny Derma=skin, Ata=characterized by) includes exclusively marine invertebrates displaying pentamerous radial symmetry and an endoskeleton of calcareous plates and spines. Jacob Klein gave the name echinodermata.
- This phylum is a collection of about 7,000 living species and constitutes some of the most beautiful members of sea fauna, such as starfishes or sea stars, sea urchins, sea cucumber, sea lilies and sand dollars etc.
- The name starfish is, however, misleading as these animals are not true fishes. A more suitable name suggested for them is sea star.
- Common genera of sea stars are Asterias, Pentaceros, Astropecten, Solaster, Heliaster, Luidia, Asterina etc. Genus Asterias includes about 150 species of which most common is *A. rubens*.
- The echinoderms are important both biologically and geologically. Biologically, there are few other groupings so abundant in the biotic desert of the deep sea, as well as shallower oceans.
- Echinoderms generally have remarkable powers of regeneration of tissue, organs, limbs, and of asexual reproduction, and in some cases, complete regeneration from a single limb.
- Geologically, the value of echinoderms is in their ossified skeletons, which are major contributors to many limestone formations, and can provide valuable clues as to the geological environment.

STAR FISH



Dorsal view



Ventral view

STAR FISH

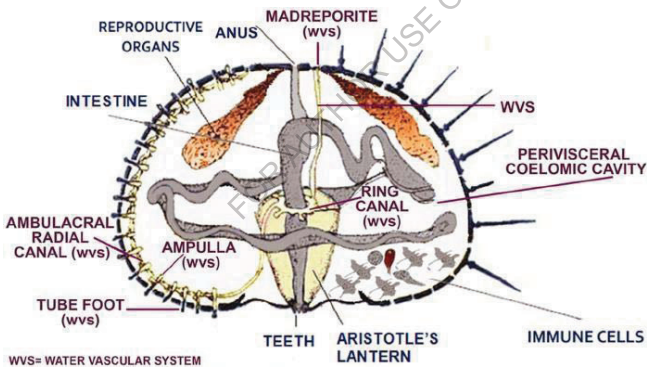
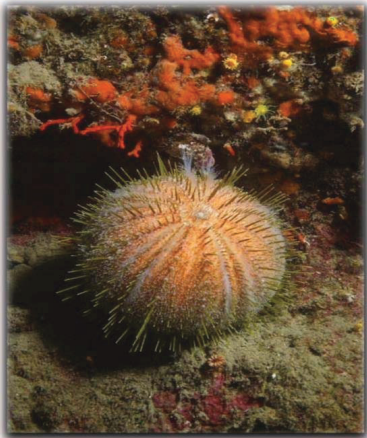
SCIENTIFIC CLASSIFICATION:

Phylum	: Echinodermata
Class	: Asteroidea
Order	: Forcipulata
Genus	: <i>Asterius rubens</i> (Star Fish)

COMMENTS:

1. Asterias is commonly known as starfish or sea star.
2. Average size of Asterias ranges from 10 to 25 cm in diameter. The body has a central disc and five arms.
3. It is Radially symmetrical animal.
4. The body is star shaped, consisting of a central disc with five radiating arms which are broad at their base and tapering towards their extremities.
5. The body consists of two surfaces, namely Aboral surface and oral surface.
6. The oral surface bears the mouth and five Ambulacral grooves with water vascular system. The tube feet occur in each Ambulacral groove.
7. Aboral surface bears a number of short, stout spines arranged in irregular rows, numerous dermal branchiate among spines and a smaller aperture, the anus which is situated near the center.
8. Pedicellariae are very small, microscopic bodies scattered all over the body.
9. Sexes are separate. Fertilization is external.
10. Development includes a free swimming bipinnaria larva.

SEA –URCHIN (*Echinus*)



SEA –URCHIN (*Echinus*)

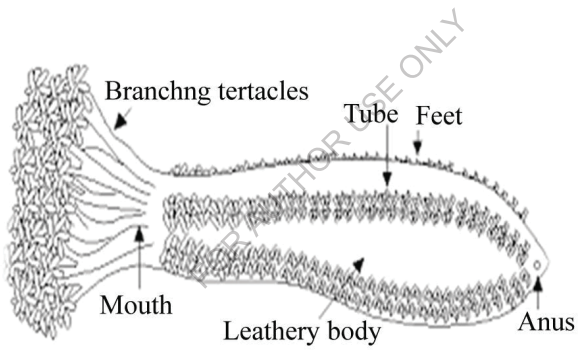
SCIENTIFIC CLASSIFICATION:

Phylum	Echinodermata
Class	Echinoidea
Order	Camarodonta
Genus	<i>Echinus</i>

COMMENTS:

1. Echinus is commonly called sea-urchin
2. It has a globe-shaped body enclosed within a shell or corona and having very long movable spines. The surface of corona is divided into five Ambulacral regions alternating with five interambulacral regions. In each Ambulacral region, there are two rows of tube feet.
3. Mouth is present on oral surface and is surrounded by peristomial. The chewing apparatus or Aristotle's lantern projects from the mouth. Anus is surrounded by a periproct and is present in the centre of the Aboral surface.
4. Water-vascular system is well developed.
5. Development is indirect and involves free swimming Echinopluteus larva.
6. Sexes are separate. Gonads are five large masses.
7. Echinus is marine, found in the sea in the rocky places.
8. Echinus is widely distributed in the Atlantic, Mediterranean and Pacific Oceans

SEA- CUCUMBER (*holothuria*)



SEA- CUCUMBER (*holothuria*)

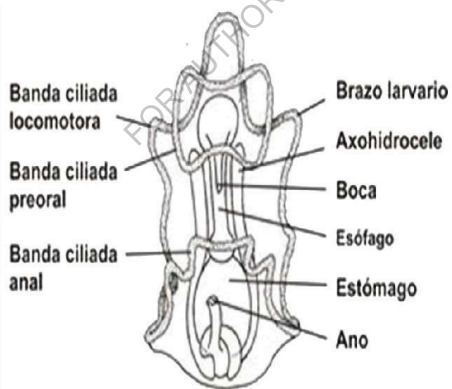
SCIENTIFIC CLASSIFICATION:

Phylum	Echinodermata
Class	Holothuroidea
Order	Aspidochirota
Genus	<i>Holothuria</i>

COMMENTS:

1. *Holothuria* is commonly known as sea-cucumber.
2. It is found in nearly every marine environment, but is most diverse on tropical shallow-water coral reefs.
3. The body is sausage-shaped, elongated, covered with leathery skin and having well developed respiratory tree.
4. Mouth is anteriorly placed, surrounded by 15-30 peltate tentacles.
5. Sexes are separate.
6. Development includes two larval stages doliolaria and auricularia respectively.
7. Holothuria feeds by pushing sand containing organic food into mouth with the help of tentacles.
8. Holothuria is found in shallow tropical and subtropical waters of Indo-Pacific area, lying together in thousands on the sandy bottom, fully exposed or somewhat hidden under plants or shell bits, etc.
9. Holothuria is found distributed in India, West Indies and Florida.

BIPINNARIA LARVA



BIPINNARIA LARVA

COMMENTS:

1. It is the second larva of star fish. It develop from Dipleurula larva.
2. It is minute and microscopic. This is the whole mount of Bipinnaria larva characteristic of the class Asteroidea.
3. After gastrula the egg develops into this free swimming ciliated larva.
4. It has one pre oral and one Postoral Ciliary band.
5. The pre oral lobe is surrounded by pre oral ciliated band.
6. The postoral ciliated band forms a complete ring between the mouth and anus.
7. The prolongations of the body are called arms. They are two posterolateral, two postoral, two Posterodorsal, two Anaterodorsal, two pre oral, one ventro median and one dorso median arms.
8. Alimentary canal is differentiated into mouth, esophagus, stomach, intestine and anus.
9. The larva feeds on diatoms.

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PROCHORDATA

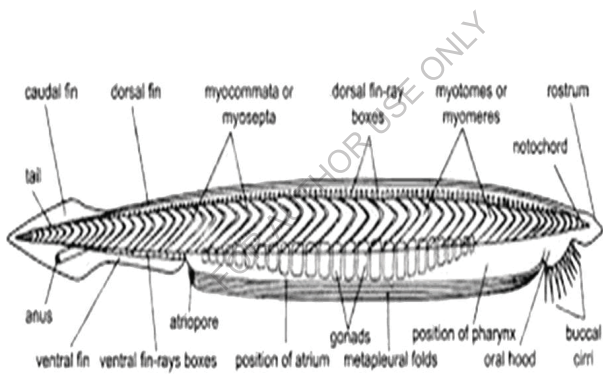
INTRODUCTION:

Protochordate, any member of either of two invertebrate subphyla of the phylum Chordata: the Tunicata (sea squirts, salps, etc.) and the Cephalochordata (amphioxus). Like the remaining subphylum of the chordates, the Vertebrata, the protochordates have a hollow dorsal nerve cord, gill slits, and a stiff supporting rod, the notochord, the forerunner of the backbone. The protochordates differ chiefly from the vertebrates in not having a backbone. Recent protochordates are thought to have evolved from the same ancestral stock as that which gave rise to the vertebrates.

Two main theories have gained general acceptance as to how the vertebrates may have evolved. One theory proposes that the ancestral form was sessile (attached), perhaps like a pterobranch but with an unspecialized larva. This larva adapted to an independent pelagic life and became sexually mature. Subsequently, the sessile stage was lost, and the vertebrates evolved from this free-swimming animal. The other, more recent theory postulates that the chordates evolved from a small fossil group called the mitrates.

For more information on protochordate groups, *see* Amphioxus ; Salp; Sea squirt; Tunicate.

AMPHIOXUS (*Branchiostoma lanceolatum*)



AMPHIOXUS (*Branchiostoma lanceolatum*)

SCIENTIFIC CLASSIFICATION:

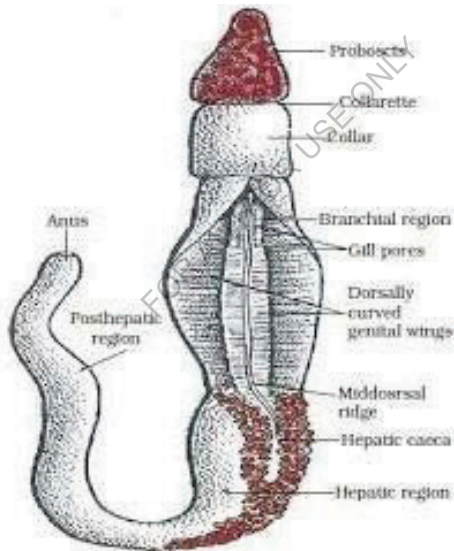
Phylum	: Chordata
Subphylum	: Cephalochordata
Class	: Leptocardii
Order	: Amphioxus

COMMENTS:

1. Amphioxus is found in shallow sub tidal waters (marine) with sandy bottoms. They reside in subtropical and temperate regions. Lancelets burrow their posterior end in the sand, with just their head exposed.
2. Branchiostoma is about 3.5 to 6.0 cm in length. The body is slender, somewhat translucent; superficially fish- like, laterally compressed and pointed at both ends, hence, the name lancelet.
3. The anterior two-thirds of the body is roughly triangular in transverse section. The posterior third is nearly oval in section. The pointed anterior end of trunk projects in front as a snout or rostrum. Below the rostrum is an oral hood formed by dorsal and lateral projections of the body. It bears twenty or more stiff oral cirri or tentacles. The body is distinguished in two regions- the trunk and tail. The true head is lacking.
4. Branchiostoma has three unpaired fins- dorsal, caudal and ventral. Paired fins are absent. Along the mid-dorsal side of the body is a dorsal fin running along the whole length. It is joined to a somewhat broader caudal fin around the tail. Mid-ventrally is a ventral fin running from the caudal fin to an Atriopore.
5. The dorsal fin is supported by a series of fin rays, and nerves the ventral fin by a series of paired fin rays. The fin rays are made of a gelatinous substance and connective tissue. Fin rays are lacking in caudal fin.
6. The oral hood is continuous Posteriorly with two lateral membranous hollow Metapleural folds running ventrally up to the atriopore.
7. The body wall shows Metameric segmentation, being made of < shaped blocks of muscles called Myotomes or Myomeres. These are found just beneath the transparent body wall. Beneath the Myotomes on both sides are present a series of gonads between mouth and Atriopore. The mouth surrounded by a frill-like membrane, the oral hood. A short distance from the posterior extremity of the body (trunk) is the anus, placed asymmetrically on the left side at the base of the caudal fin. The post-anal portion of the body is called the tail.

HEMICHORDATA

BALANOGLOSSUS

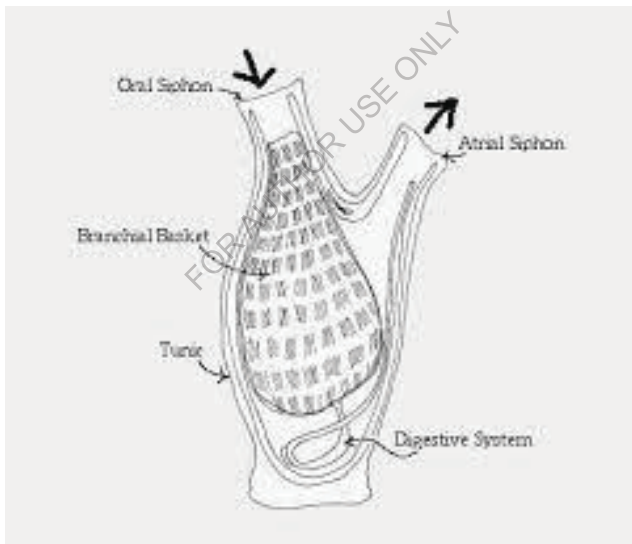


BALANOGLOSSUS

COMMENTS:

1. All Hermichordates are Marine. Some are solitary and slow moving, others are sedentary and colonial.
2. Body is stout and Un segmented, and has a worm-like or vase-like
3. Three distinct regions namely Proboscis, collar and trunk are present.
4. 4 They are bilaterally symmetrical and triploblastic.
5. They have organ-system level of organization.
6. There are no Locomotory appendages. The collar may bear arms and tentacles.
7. Body wall consists of single layered epidermis and musculature of smooth longitudinal fibers.
8. Their body consists of a true coelom with 3 parts corresponding to the 3 body divisions: an unpaired proboscis coelom, a paired collar coelom and a paired trunk coelom.
9. Digestive tract is complete. Proboscis contains a hollow out growth from the gut, called the Buccal diverticulum or Stomochord and was regarded as notochord in the past.
10. All feed on microorganisms and debris by filtering or ciliary mechanism.
11. 10. Respiration occurs by a pair to numerous pairs of gill slits or through the general body surface.
12. Circulatory system includes a dorsal heart, two main longitudinal vessels; a dorsal and a ventral, interconnected by small lateral vessels and sinuses.
13. 12 .Blood is colorless and without corpuscles.
14. Nervous system is diffuse consisting of an epidermal plexus of nerve cells and nerve fibers.
15. Excretory system comprises of a proboscis gland, or glomerulus, situated in the proboscis and connected with the blood vessels.
16. Sexes are separate or united. The gonads may be in several pairs or only one in pair.

2. UROCHORDATA (or) TUNICATA:



ASCIDIAN (Herdmania)

COMMENTS:

1. They are marine, mostly sessile, filter-feeders.
2. Body is enclosed in a leathery test or tunic sheath, composed of tunicin (cellulose) so called tunicates.
3. The notochord occurs only in the tail of larva and disappears in the adults called retrogressive metamorphosis.
4. The nerve chord (neural tube) is present in the larva, but is replaced by a single dorsal ganglion in the adult.
5. The gill slits or stigma are numerous, persist in the adults and open into the atrium, instead of opening to the exterior. There are no true gills but are called bronchial basket.
6. Circulatory system is of open type. Blood consists of Venadocytes.
7. Excretory system is lacking.
8. Asexual reproduction occurs by budding.

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AGNATHA

INTRODUCTION

- Agnatha- ("no jaws") is a super class of jawless fish in the phylum Chordata, subphylum Vertebrata, consisting of both present (cyclostomes) and extinct (condones and ostracoderms) species All living and Most extinct.
- Agnatha do not have an identifiable stomach or any appendages. The oldest fossil agnathans appeared in the Cambrian, and two groups still survive today: the lampreys and the hagfish, comprising about 120 species in total.
- Hagfish are considered members of the subphylum Vertebrata, because they secondarily lost vertebrae; before this event was inferred from molecular and developmental data, the group Craniates was created by Linnaeus (and is still sometimes used as a strictly morphological descriptor) to reference hagfish plus vertebrates.
- In addition to the absence of jaws, modern Agnathans are characterized by absence of paired fins; the presence of a notochord both in larvae and adults; and seven or more paired gill pouches. Lampreys have a light sensitive pineal eye (homologous to the pineal gland in mammals).
- All living and most extinct Agnatha do not have an identifiable stomach or any appendages. Fertilization and development are both external. There is no parental care in the Agnatha class. The Agnatha are Ectothermic or cold blooded, with a cartilaginous skeleton, and the heart contains 2 chambers.
- While a few scientists still regard the living Agnathans as only superficially similar, and argue that many of these similarities are probably shared basal characteristics of ancient vertebrates, recent classification clearly place hagfish (the Myxin or Hyperotreti) with the lampreys (Hyperoartii) as being more closely related to each other than either is to the jawed fishes.

PETROMYZON



PETROMYZON

SCIENTIFIC CLASSIFICATION:

Kingdom: Animalia

Phylum : Chordata

Class: Hyperoartia

Order: Petromyzontiformes

Genus: *Petromyzon*

Species: *P. marinus*

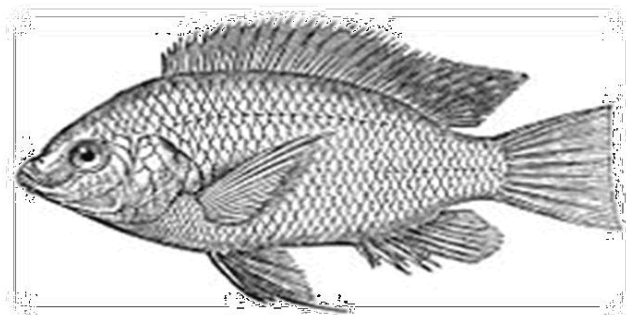
COMMENTS:

- ***Petromyzon*** lack jaws, retain a notochord throughout their life, have vertebral column composed of cartilage that surrounds notochord, looks like an eel.
- Body Plan: Bilateral symmetry, has eye, gill, and a tail
- Feeding/Nutrition: In their larva phase, they eat algae and small organisms. In their parasitic phase, they feed off a host (fish). In their spawning phase, they don't eat.
- Reproduction: The sperm and eggs enter the body cavity and are then discharged via the paired genital pores, which open into the urogenital sinus, and then exit through the urogenital aperture.
- Life Cycle: The adults spawn in cold, clear streams. They spawn, and then die shortly after which is called semelparity. The eggs hatch into larvae (ammocoetes). These larvae are filter feeders and live in small mud burrows in the stream for up to 10 years. They eventually undergo metamorphosis. This phases causes their bodily structure and function to change. This is when they grow rapidly and become sexually mature.
- Locomotion: Lampreys are able to move because they have a notochord. A notochord helps support the body and also helps them move.
- Notable Physical Features: On the outside of a Lamprey, you can see its eye, bilaterally symmetrical body, anterior dorsal fin, external gill slits, posterior dorsal fin, and the caudal fin. Behavior: Lampreys don't have a distinct character. They mostly mind their own business.
- Defense/Survival Strategies: Lampreys use their blood suckers to kill fish. Some people have used chemicals and dams to prevent Lampreys from getting to their fish farms because they are so harmful.
- Behavior: Lampreys don't have a distinct character. They mostly mind their own business.
- Defense/Survival Strategies: Lampreys use their blood suckers to kill fish. Some people have used chemicals and dams to prevent Lampreys from getting to their fish farms because they are so harmful.

TILAPIA (*Oreochromis mossambicus*)



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TILAPIA (*Oreochromis mossambicus*)

SCIENTIFIC CLASSIFICATION:

Kingdom : Animalia

Phylum : Chordata

Class : Actinopterygii

Order : Cichliformes

Family : Cichlidae

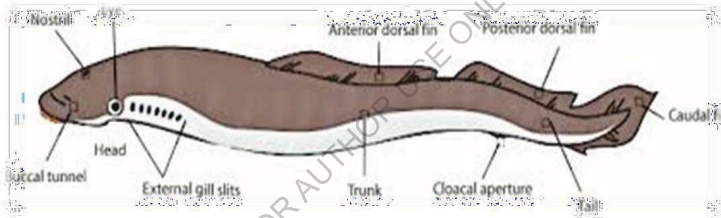
Genus : *Oreochromis*

Species : *O. mossambicus*

COMMENTS:

- Tilapia typically has laterally compressed, deep bodies. Like other cichlids, their lower pharyngeal bones are fused into a single tooth-bearing structure.
- A complex set of muscles allows the upper and lower pharyngeal bones to be used as a second set of jaws for processing food (cf. morays), allowing a division of labor between the "true jaws" (mandibles) and the "pharyngeal jaws". This means they are efficient feeders that can capture and process a wide variety of food items.
- Their mouths are protractible, usually bordered with wide and often swollen lips. The jaws have conical teeth.
- Typically, tilapia have a long dorsal fin, and a lateral line which often breaks towards the end of the dorsal fin, and starts again two or three rows of scales below. Some Nile tilapia can grow as long as 2.0 ft.
- Other than their temperature sensitivity, tilapia exists in or can adapt to a very wide range of conditions. An extreme example is the Salton Sea, where tilapia introduced when the water was merely brackish now lives in salt concentrations so high that other marine fish cannot survive.
- Tilapia are also known to be a mouth-breeding species, which means they carry the fertilized eggs and young fish in their mouths for several days after the yolk sac is absorbed.

EEL



EEL

SCIENTIFIC CLASSIFICATION:

Kingdom : Animalia

Phylum : Chordata

Class : Actinopterygii

Order : Anguilliformes

Genus : *Anguilla*

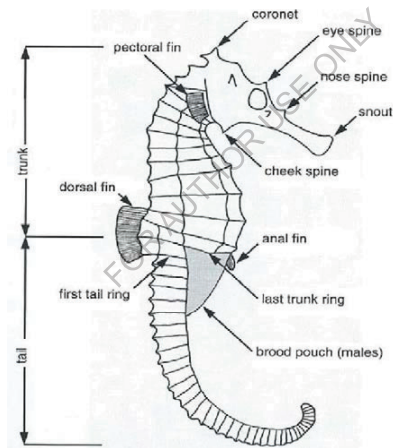
Species : *A. bengalensis*

COMMENTS:

- Behavior: Lampreys don't have a distinct character.
- They mostly mind their own business.
- Defense/Survival Strategies: Lampreys use their blood suckers to kill fish.
- Some people have used chemicals and dams to prevent Lampreys from getting to their fish farms because they are so harmful

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HIPPOCAMPUS



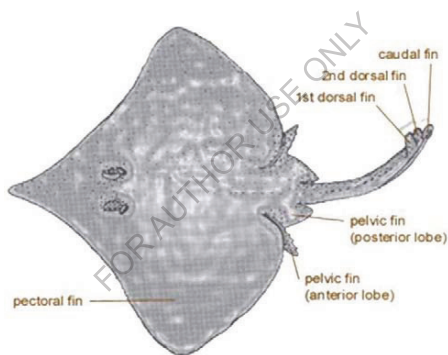
HIPPOCAMPUS

COMMENTS:

1. Body supported by hard bony plates with spines and tubercles
2. 10 to 12 Exo skeletal rings present in the trunk.
3. Head bent at right angle to the trunk.
4. Tail longer than trunk and prehensile.
5. Presence of a tubular snout.
6. Mouth at the tip of the snout.
7. Bronchial aperture dorsal.
8. Eyes shifted dorsally.

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NARCINE



NARCINE

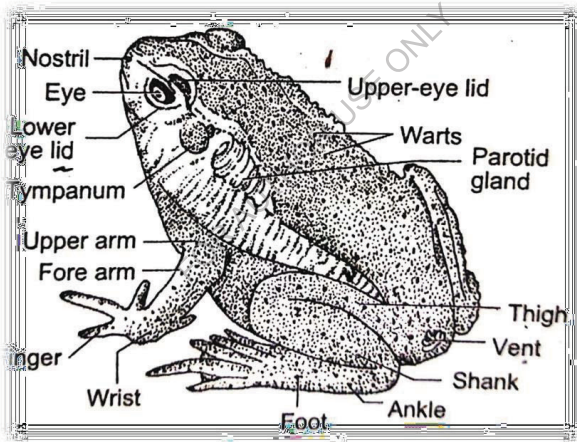
SCIENTIFIC CLASSIFICATION:

Phylum	Chordata
Subphylum	Vertebrata
Superclass	Gnathostomata
Class	Elasmobranchii
Subclass	Selachi
Superorder	Hypotremata

COMMENTS:

1. Presence of electric organs between head and pectoral fins.
2. Preorbital cartilages greatly reduced.
3. Body sub circular Anteriorly.
4. Nasofrontal lobe quadrangular.
5. Pectoral fins large and fused with the head and body.
6. Pelvic fins fairly large.
7. Spiracles present one pair, one on either side between the head and the pectoral fin.
8. Length up to 60 cm. 5 large prominent eye spots on the brownish back.
9. Maxilla free from preopercular.
10. Clavicle absent.
11. Spiracle absent.
12. Swim bladder present.

BUFO



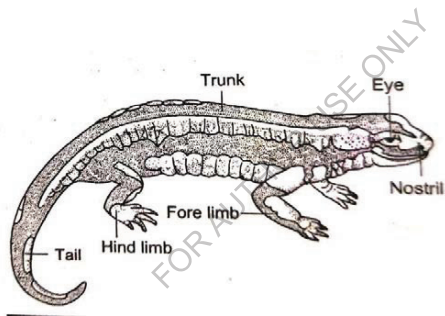
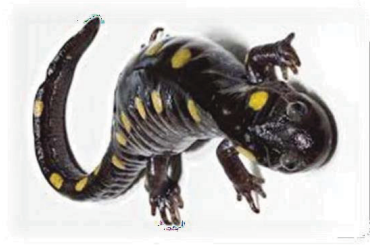
BUFO

COMMENTS:

- Bufo is cosmopolitan except Australia.
- Indian toad is nocturnal, found in damp and dark places, does not visit water except during breeding season.
- Body ovoid, flattened, size 8-11 cm, skin rough dry and warty with brown spotted dorsal surface and pale or whitish ventral surface.
- It lacks teeth in its Buccal cavity, tongue sticky but not bifurcated.
- A pair of large parotid glands situated behind the tympana, to produce milky poisonous fluid.
- Male toad has a single vocal sac but develops amplexusory pads on two inner fingers of each hand.
- The web is rudimentary.
- 2 Euryhaline frogs, found in mangrove – dwellings are *Rana cancrivara* & *Bufo viridis*.
- 2 Neoteny: Retention of a larval or Ambryonic trait in the adult body. Eg. *Necturus*, *Proteus*, *Siren*.

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SALAMANDER



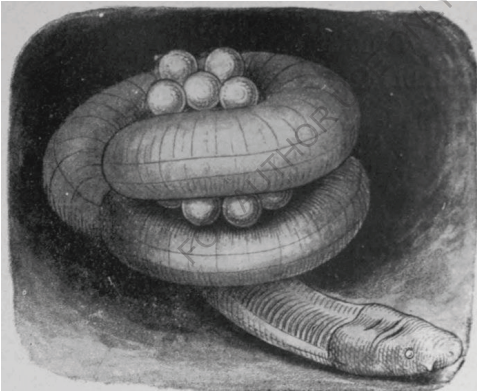
SALAMANDER

COMMENTS:

- Salamander is found in Europe.
- It is a semi terrestrial, lizard like Amphibian, found in crevices and below logs and stones.
- It has shining black skin with yellow spots.
- Tail sub cylindrical, gills or gill slits are absent.
- It is ovoviviparous; eggs are retained and developed inside the body of female who gives birth to larvae which complete their development in water.

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ICHTHIOPHIS



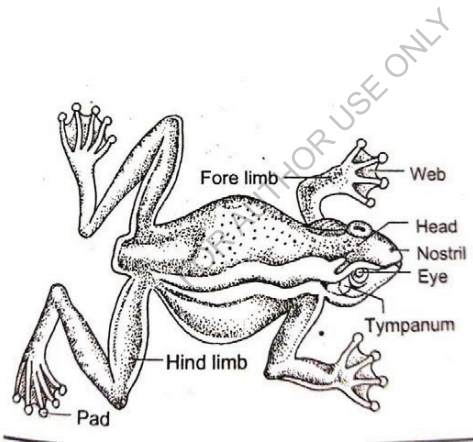
ICHTHIOPHIS

COMMENTS:

- It is limbless amphibian of 15-22 cm length, found underground in India, Ceylon, Malaya and Siam.
- Head small, eyes microscopic and covered with opaque skin (that is why it is blind worm), sensory tentacles occur between nasal and eyes, they are meant for tasting food, no middle ear and tympanum.
- No limbs, no girdles, anus sub terminal so that there is almost no tail.
- Dermal minute scales are embedded in skin.
- Males have eversible cloacae which act as copulatory organ. Fertilization is internal.
- Females guard the eggs by coiling round them. The young ones hatch out only when the burrows get flooded with water.

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RHACOPHORUS (Flying frog)



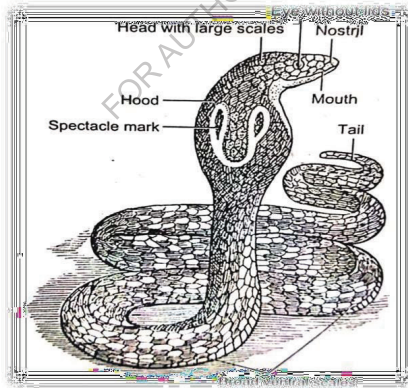
RHACOPHORUS (Flying frog)

COMMENTS:

- It is found in Africa and South Eastern Asia.
- It possesses characteristic large webs developed between the much elongated digits. Webs and flattened body serve as a parachute in gliding from higher elevation to lower ones, so it is designated as Flying frog.

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REPTILIA (Cobra)

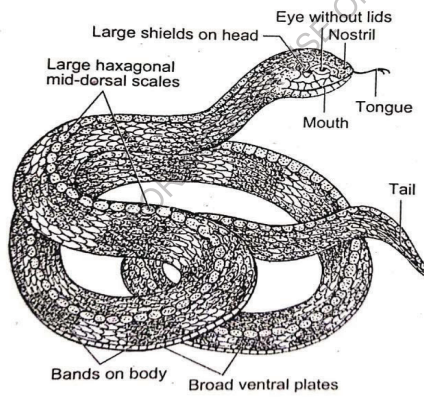


REPTILIA (Cobra)

COMMENTS:

- Naja is most common and deadly or poisonous snake of India. It lives in burrows, undertones, in thick vegetation and may also come in human dwellings.
- It grows up to 6-8 meters.
- It is brown or brownish black. It has small and indistinct head. The neck can dilate (due to rising of cervical ribs) and form a hood. It may or may not bear spectacle marks dorsally.
- The pupil around, the 3rd supralabial shield of upper lip touches eye and nasal shield. The sub caudal shields are arranged in 2 rows.
- The fangs are anterior, grooved and permanently erect.
- When annoyed, it raises the front part of the body, spreads its hoods and hisses through the nose. It bites with a chewing motion making several wounds.
- The venom is Neurotoxin and fatal being more deadly than that of vipers.
- It feeds on small birds, rats, frogs lizards etc.
- Herpestes (Mongoose) kills it.
- Ophiophagus Hannah (king Cobra) is the largest and the deadliest venomous snake. Naja Hannah, Hamadryads and Naja bungarus are found in India. Malaya and Southern China. It grows up to 18 feet, lives in dense forests. It feeds on non-poisonous snakes, monitor lizards etc. It is the only snake that builds nest. It is oviparous.

BUNGARUS (KRAIT)

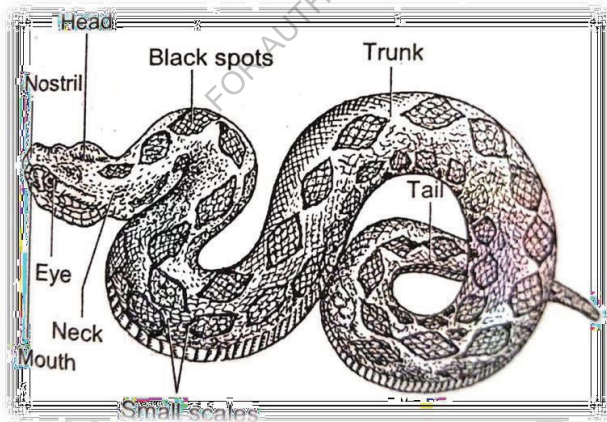


BUNGARUS (KRAIT)

COMMENTS:

- It occurs all over India and Malaya.
- It is about 3 feet long, slender, brown coloured and has darker bluish black markings with white cross bars.
- Vertebral are large and hexagonal. The 4th infralabial is the largest. The sub caudal is single. The head is covered with large shields.
- Fangs are small. The wounds inflicted on victim are superficial but poison is 3 times more virulent or deadly than that of cobra.
- It is terrestrial, diurnal, feeding on small amphibians, lizards, snakes and mammals.
- Bungarus fasciatus is the banded krait, restricted to Northern India and Indonesia. It has black and yellow sub equal cross bars and the back is keeled.

PIT VIPERS



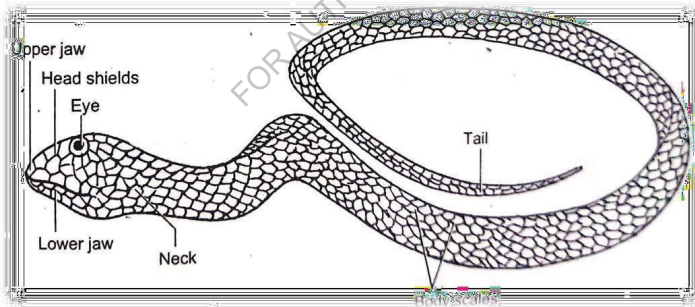
PIT VIPERS

COMMENTS:

- It is commonly called Rattle snake, found mainly in America and Mexico, grows up to 8 feet and has a prominent sensory pit between the eye and nostril.
- Dorsal surface of head has small scales.
- Fangs are long, movable and folded back when not in use.
- At the end of tail is a rattle having a small hard button at the tip and usually 10-12 dry horny rings, formed as hollow segments by the end of the courted skin being left attached to the tail. This rattle is always headed vertically and it makes a grating sound during locomotion and is probably for warning.
- The Rattle snake is highly poisonous, carnivorous and viviparous.

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DRYOPHIS



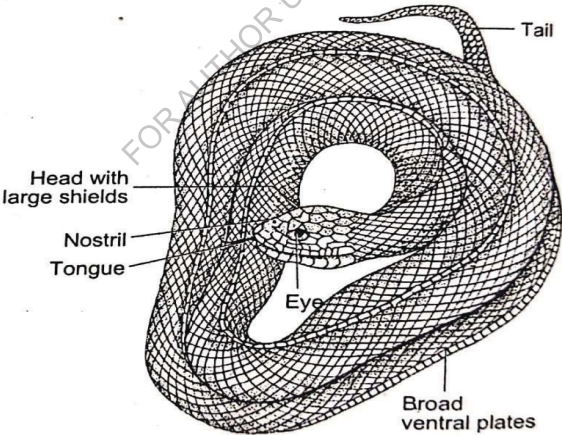
DRYOPHIS

COMMENTS:

- It is whip-snake with excessively slender body like the cord of a whip and a prehensile tail.
- Head is narrow with tapering snout and horizontal pupils.
- It is viviparous and feeds on small reptiles and birds.
- Its saliva is weakly poisonous.

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PTYAS



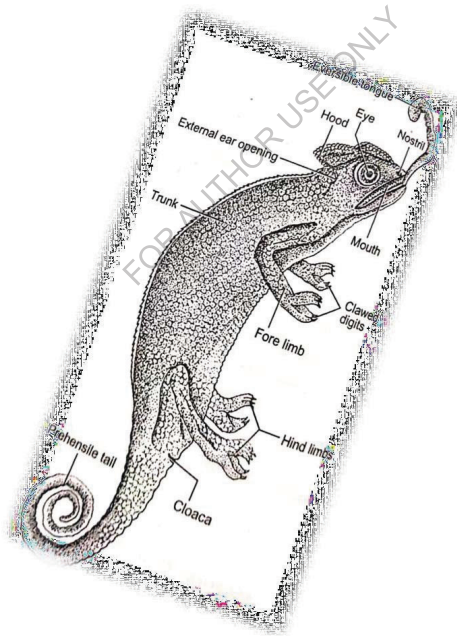
PTYAS

COMMENTS:

- Zamenis mucosus is commonly known as Dhaman or Rat-snake of India. (Also found in Burma and Java)
- It feeds on rats, small rodents, frogs, birds and often enters human dwellings in search of rats or mice.
- It is 8 feet long with elongated tail, olive green dorsally and whitish ventrally. The scales are smooth, eyes large with round pupil, sharp ridge along the backbone.
- It is highly active and alert, very swift, ill-tempered, bites viciously but is non poisonous.
- It is oviparous.
- It is regarded as friend of farmers as it destroys rats.

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CHAMELEON ZEYIANICUS – GIRGIT



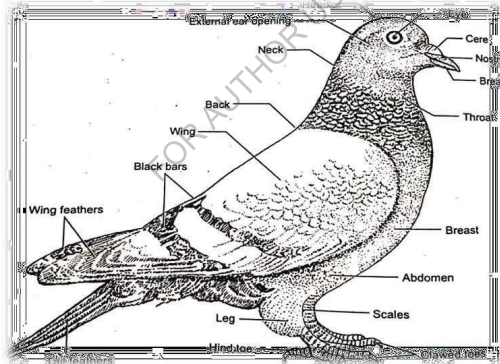
CHAMELEON ZEYIANICUS – GIRGIT

COMMENTS:

- It is highly specialized for arboreal life and is famous for changing its colour rapidly to blend with its surroundings. The phenomenon is called metachrosis. Its normal colour is greenish black.
- Body is 35-45 cm, long and laterally compressed. The skin is granulated. The head bears a prominent helmet or hood consisting of squamosal and occipital bones.
- Eyes are large with narrow pupil. The eyelids are fused together.
- Tympanum is absent. The tongue is very long and eversible, and bears adhesive bulb to capture insects.
- Feet zygodactylus (digits arranged into two groups of 2+3) to grasp branches firmly.
- Tail is long and prehensile.
- It is ovoviviparous.

BIRDS

PIGEON

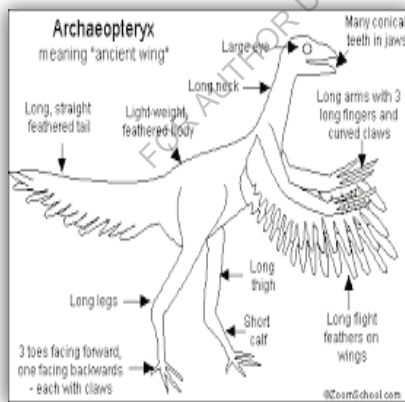


PIGEON

COMMENTS:

- **Pigeon**, any of several hundred species of birds constituting the family Columbidae (order Columbiformes). Smaller forms are usually called doves, larger forms pigeons. An exception is the white domestic pigeon, the symbol known as the “Dove of Peace.”
- Pigeons occur worldwide except in the coldest regions and the most remote islands.
- All members of the family suck liquids, rather than sip and swallow as do other birds, and all pigeon parents feed their young “pigeon’s” the sloughed-off lining of the crop, the production of which is stimulated by the hormone prolactin.
- The nestling obtains this “milk” by poking its bill down the parent’s throat.
- Pigeons are gentle, plump, small-billed birds with a skin saddle (cere) between the bill and forehead.
- All pigeons strut about with a characteristic bobbing of the head. Because of their long wings and powerful flight muscles, they are strong, swift fliers.
- Pigeons are monogamous; i.e., they mate for life, and the survivor accepts a new mate only slowly.
- The female lays two glossy white eggs in a flimsy nest that barely holds them. The female generally incubates the
- Eggs by night, the male by day. The incubation period is 14 to 19 days, but the young are cared for in the nest for another 12 to 18 days.
- The identity of the physical structure within the pigeon’s body that collects this magnetic field information and sends it to the brain remains unknown, but some scientists suspect that it may lie within the inner ear.

ARCHAEOPTERYX



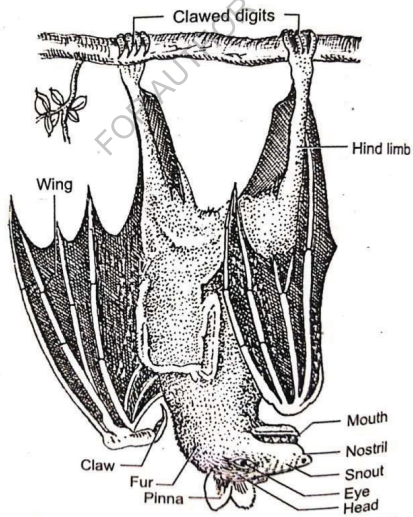
ARCHAEOPTERYX

COMMENTS:

1. These features make Archaeopteryx a clear candidate for a transitional fossil between non-avian dinosaurs and birds. Thus, Archaeopteryx plays an important role, not only in the study of the origin of birds, but in the study of dinosaurs.
2. Most of these eleven fossils include impressions of feathers. Because these feathers are of an advanced form (flight feathers), these fossils are evidence that the evolution of feathers began before the Late Jurassic.
3. The type specimen of Archaeopteryx was discovered just two years after Charles Darwin published *On the Origin of Species*. Archaeopteryx seemed to confirm, Darwin's theories and has since become a key piece of evidence for the origin of birds, the transitional fossils debate, and confirmation of evolution.
4. Archaeopteryx plays an important role, not only in the study of the origin of birds, but in the study of dinosaurs. It was named from a single feather in 1861.[8] That same year, the first complete specimen of Archaeopteryx was announced.
5. The type specimen of Archaeopteryx was discovered just two years after Charles Darwin published *On the Origin of Species*.
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MAMMALS

BAT



BAT

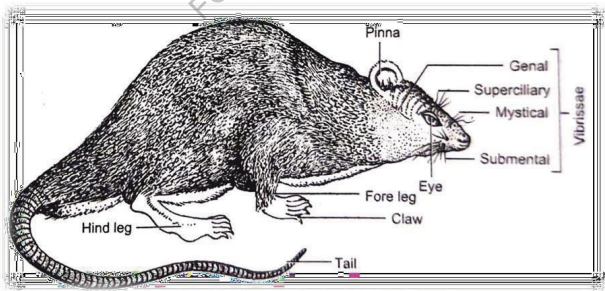
SCIENTIFIC CLASSIFICATION:

Kingdo	Animali
Phylu	Chordat
Class	Mammali
<i>Clade</i>	Scrotifer
Order	Chiropter

COMMENTS:

- **Bats** are mammals of the order **Chiroptera**; with their forelimbs adapted as wings, they are the only mammals naturally capable of true and sustained flight.
- Bats are more maneuverable than birds, flying with their very long spread-out digits covered with a thin membrane or patagium.
- The second largest order of mammals after rodents, bats comprise about 20% of all classified mammal species worldwide, with over 1,200 species.
- These were traditionally divided into two suborders: the largely fruit-eating mega bats, and the echo locating micro bats. But more recent evidence has supported dividing the order into Yinpterochiroptera and Yangochiroptera, with mega bats as members of the former along with several species of micro bats.
- Many bats are insectivores, and most of the rest are frugivores (fruit-eaters). A few species feed on animals other than insects; for example, the vampire bats feed on blood.
- Most bats are nocturnal, and many roost in caves or other refuges; it is uncertain whether bats have these behaviors to escape predators. Bats are present throughout the world, with the exception of extremely cold regions.
- They are important in their ecosystems for pollinating flowers and dispersing seeds; many tropical plants depend entirely on bats for these services.
- Bats provide humans with some benefits, at the cost of some threats. Bat dung has been mined as guano from caves and used as fertilizer.
- Bats consume insect pests, reducing the need for pesticides. They are sometimes numerous enough to serve as tourist attractions, and are used as food across Asia and the Pacific Rim.
- They are natural reservoirs of many pathogens, such as rabies; and since they are highly mobile, social, and long-lived, they can readily spread disease.
- In many cultures, bats are popularly associated with darkness, malevolence, witchcraft.

RAT



RAT

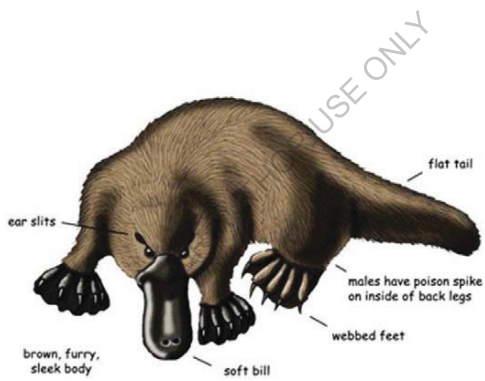
SCIENTIFIC CLASSIFICATION:

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
(unranked):	Glires
Order:	Rodentia

COMMENTS:

- **Rats** are various medium-sized, long-tailed rodents.
- Species of rats are found throughout the order Rodentia, but stereotypical rats are found in the genus *Rattus*.
- Other rat genera include *Neotoma* (pack rats), *Bandicota* (bandicoot rats) and *Dipodomys* (kangaroo rats).
- Rats are typically distinguished from mice by their size.
- Generally, when someone discovers a large muroid rodent, its common name includes the term *rat*, while if it is smaller; its name includes the term mouse.
- The common terms *rat* and *mouse* are not taxonomically specific.

PLATYPUS



PLATYPUS

COMMENTS:

- a) The prototheria are described as primitive unfinished mammals by Romer.
- b) The mammary glands of prototheria are modified sweat glands unlike the Eutheria. Teat less mammary glands is characteristic of prototheria.
- c) Males possess functional mammary glands. The feeding of young with milk by father and mother (parents) 'is known as gynaecomastism.
- d) Auditory Pinna or external ear is absent.
- e) The skull is without tympanic bulla and lacrimals. Vertebrae are without epiphyses.) Cervical vertebrae bear ribs. Ribs are single headed having capitulum only. A large coracoid with precoracoid and a T shaped interclavicles is present in pectoral girdle.
- f) The pelvic girdle has a pair of epipubic bones which are absent in Eutherians.
- g) The body temperature is low and ranges from 25° C to 28°C. Hence, they are described as heterothermous mammals. i) A cloaca is present into which ureters and urinogenital ducts open. Oviducts remain separate throughout the length and open into the cloaca.
- h) Testes are abdominal. Penis helps to pass out the Females are oviparous laying large eggs with much amount of yolk (megalecithal).
- i) Optic lobes are 4 in numbered.
- j) A cloaca is present into which ureters and urinogenital ducts open. Oviducts remain separate throughout the length and open into the cloaca.
- k) Testes are abdominal. Penis helps to pass out the Excreta.

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